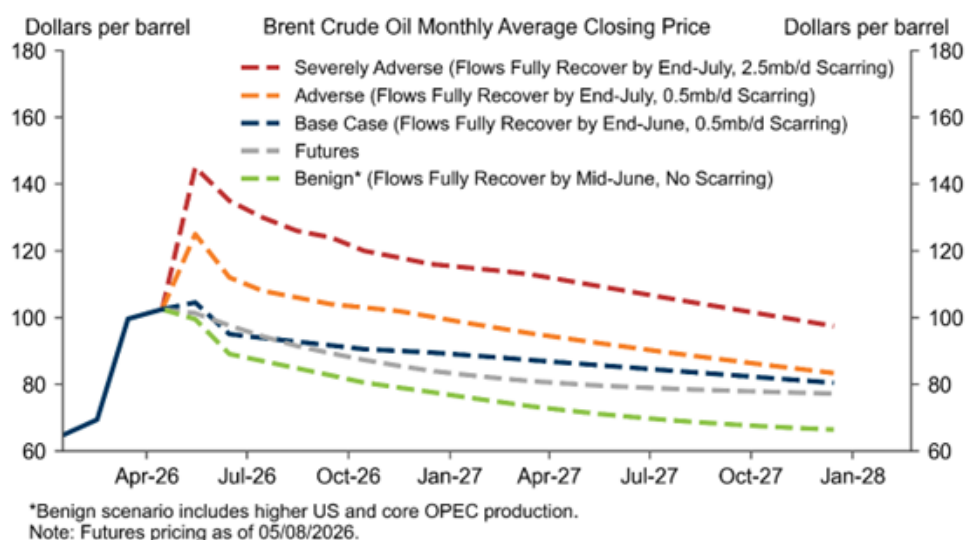


Global Views: Bending, Not Breaking

1. We see three reasons why the 10-week closure of the Strait of Hormuz has hurt global growth only moderately so far. First, oil prices have not risen as much as expected, partly because of unusually high pre-war inventories and partly because markets never lost faith that very large consumer price hikes would prompt a US policy shift. Second, physical shortages in areas such as jet fuel have so far been met with relatively painless forms of demand destruction (e.g. a large shift to renewables in China and reduced flight schedules on lower-value routes globally). Third, fiscal policy, the AI boom, and—with a brief interruption in March—financial conditions have been supportive all year. Under our baseline assumption of a gradual Strait reopening that starts soon and finishes in late June, we see Brent prices stable in the near term and edging down to \$90/barrel by year-end. However, the risks remain tilted toward more adverse outcomes, higher oil prices, and greater economic damage.

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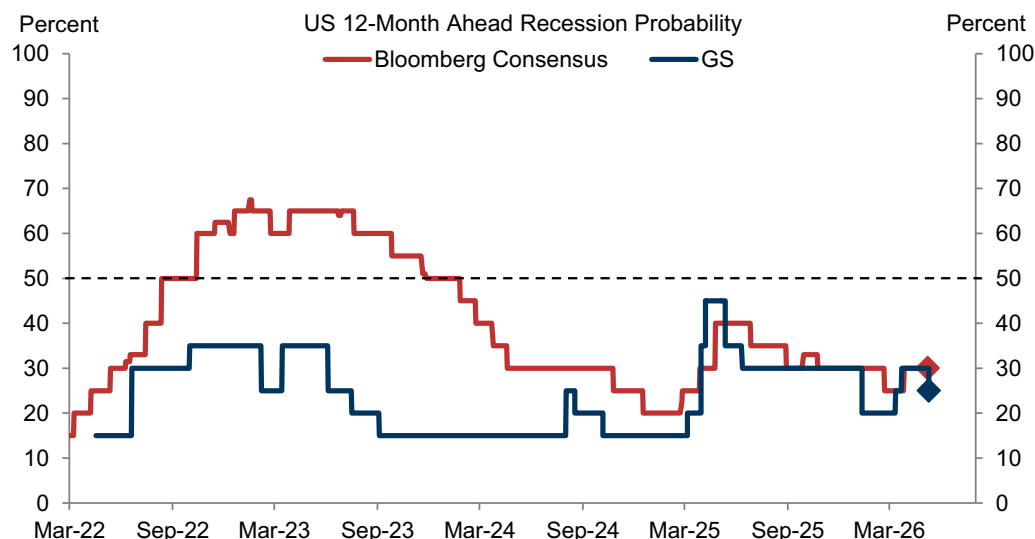
Exhibit 1: Oil Price Risks Still Skewed to the Upside



Source: Goldman Sachs Global Investment Research

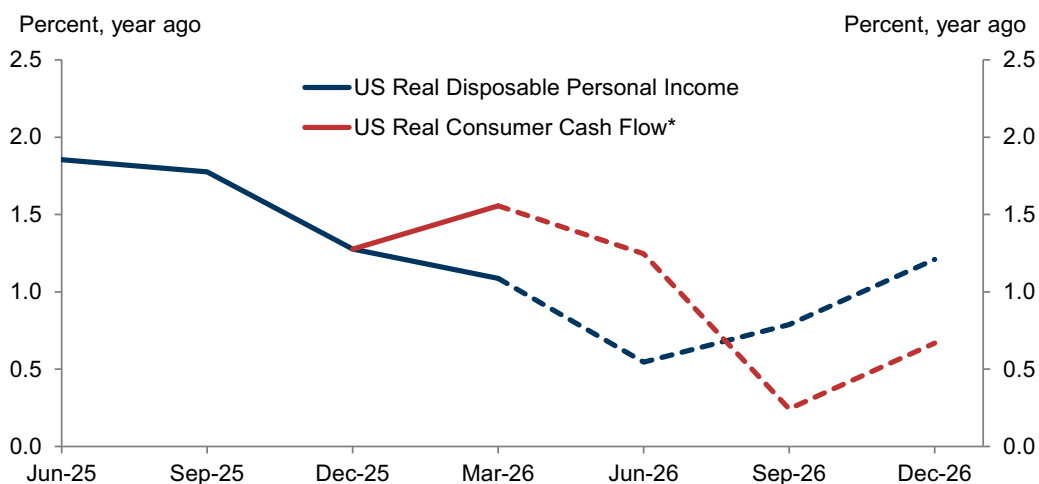
2. We have shaved our 12-month US recession probability from 30% to 25% as economic activity has held up well and our financial conditions index has eased back below pre-war levels. While headline GDP growth fell short of expectations in Q1, private domestic final sales grew at a solid 2.5% in real terms. Moreover, the labor market news has mostly beaten expectations, with a bigger-than-expected 115k increase in nonfarm payrolls in April, a stable 4.3% unemployment rate, and a sharp decline in jobless claims.

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Exhibit 2: Recession Risk Looks Incrementally Lower

Source: Bloomberg, Goldman Sachs Global Investment Research

3. But our recession risk estimate remains 5pp above the pre-war level and 10pp above the unconditional long-term average. Moreover, our baseline forecast entails below-trend growth in coming quarters, mainly because consumers face weakness in real disposable personal income and real cash flow. The boost to cash flow from strong tax refunds is about to fade, and the drags from higher gas prices, slowing nominal wage gains, and reduced healthcare subsidies are likely to persist. With the personal saving rate already down to 3.6%, the lowest level in three years, weaker income and cash flow is likely to translate into weaker consumer spending.

Exhibit 3: Real Income and Cash Flow Are Likely to Soften in H2

Note: Dashed lines indicate GS forecasts.

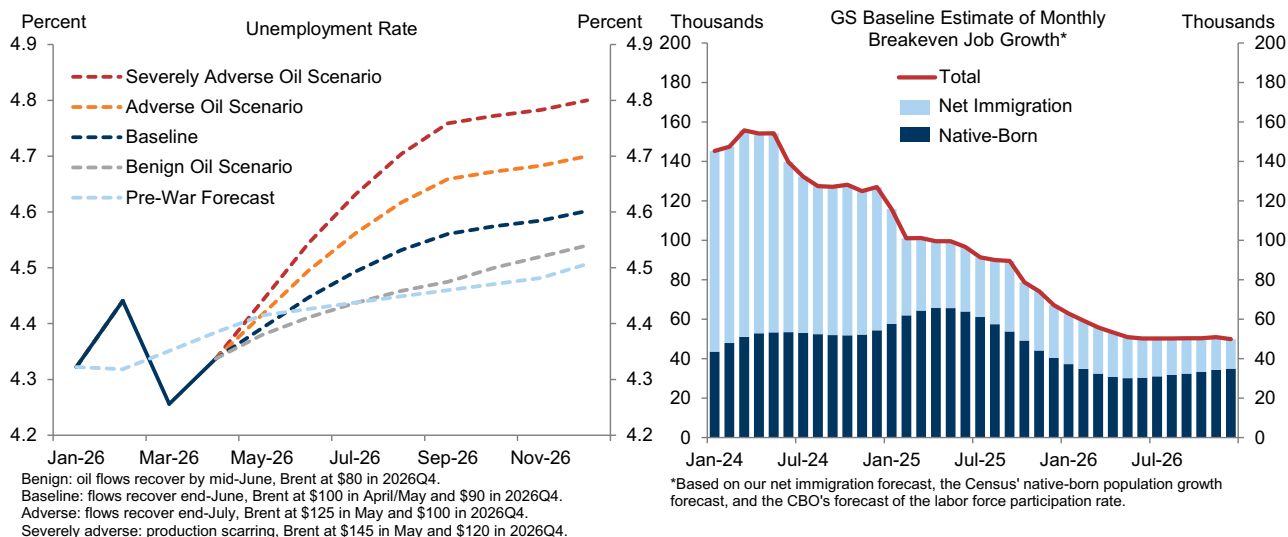
*Adjusted for the timing of OBBBA-related boosts to tax refunds and reductions in tax payments.

Source: Haver Analytics, Bureau of Economic Analysis, Goldman Sachs Global Investment Research

4. Beyond the weaker growth outlook, there are two other reasons why the US unemployment rate is likely to drift higher. First, US productivity has already accelerated from a 1.5% trend before the 2020 pandemic to 2.1% since then, and the improvement

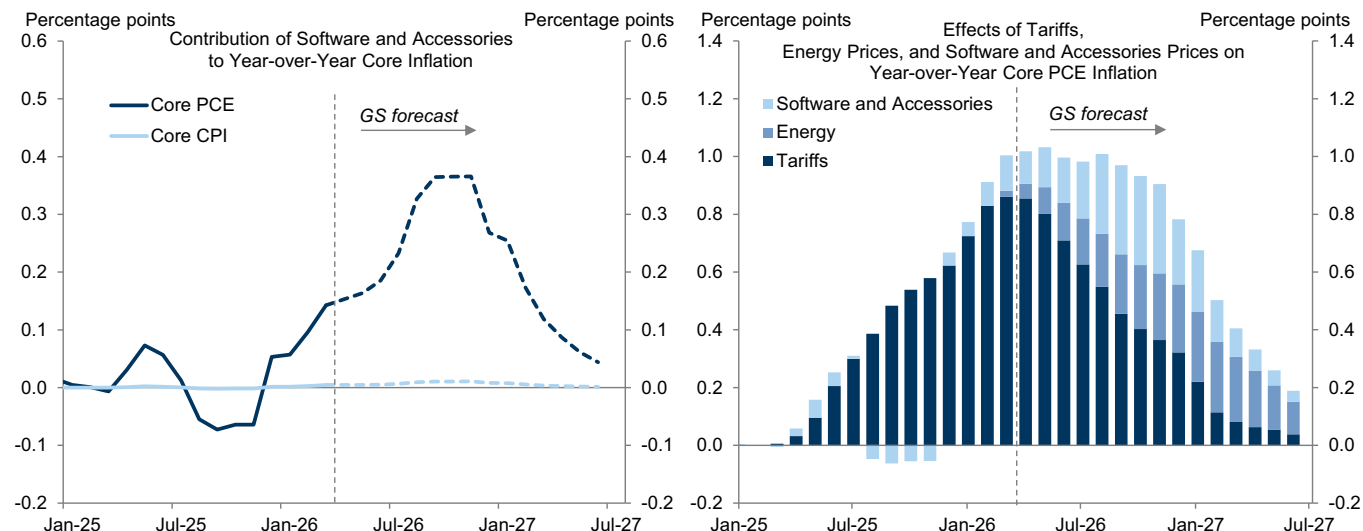
is likely to continue as the AI impact broadens. This means fewer new jobs for any given GDP increase. Second, we believe the Fed staff and many private-sector analysts have overshot in revising down their estimates of the “breakeven” rate of payroll growth to around zero. Our estimate of around 50k is consistent with the fact that the unemployment rate has fallen by only a few basis points on net this year, despite average payroll growth currently reported at 76k.

Exhibit 4: Unemployment Likely to Rise, in Part Because Breakeven Payroll Pace Remains Positive



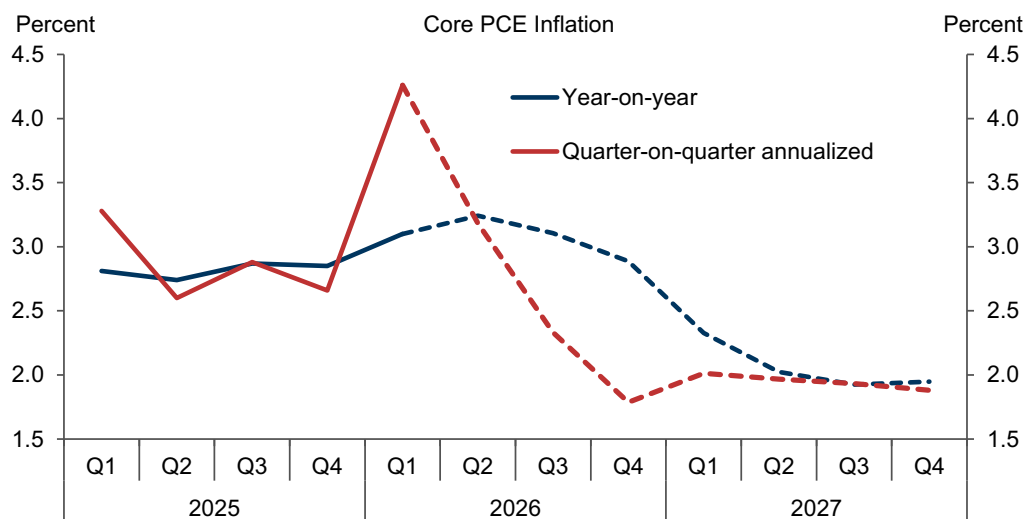
Source: Goldman Sachs Global Investment Research

5. Two distortions related to the AI boom have worsened the near-term outlook for core PCE inflation. First, after adding new AI features to software packages such as MS Office, Microsoft and other firms now charge a higher price for the new bundle than for the prior standalone package. The official statistics don't adjust for the expanded offering and thus incorrectly treat this as a pure price increase. Second, the combined weight of software and accessories—where prices have also surged on the back of costlier memory—is 30x higher in PCE than CPI, apparently because the underlying survey incorrectly captures not only consumer demand but also some demand from small businesses. Together, these distortions are likely to boost the year-on-year rate of core PCE inflation by 0.4pp later this year. Combined with an estimated 0.3pp pass-through from higher energy prices, this will delay the renewed core PCE disinflation we have been expecting.

Exhibit 5: Software Is Yet Another Transitory Inflation Shock

Source: Haver Analytics, Goldman Sachs Global Investment Research

6. The longer-term outlook for core PCE inflation is better. While the boosts from tariffs, energy, and software/accessories should fade over the next year, two key drags are likely to remain in place. First, rent inflation should continue to slow as the official CPI/PCE shelter data catch up with market rents, which remain weak because investors overbuilt apartments in the immediate post-pandemic era in several markets (especially in the Sun Belt). Second, the labor market is a source of disinflation, with nominal wage growth of 3.6% year-on-year based on our GS wage tracker and unit labor costs growing just 1.2% year-on-year in Q1. We therefore see sequential core PCE inflation returning to 2% later in 2026, though the year-on-year rate is unlikely to get back to target until mid-2027.

Exhibit 6: Core PCE Inflation Should Look Better by Year-end, Especially in Sequential Terms

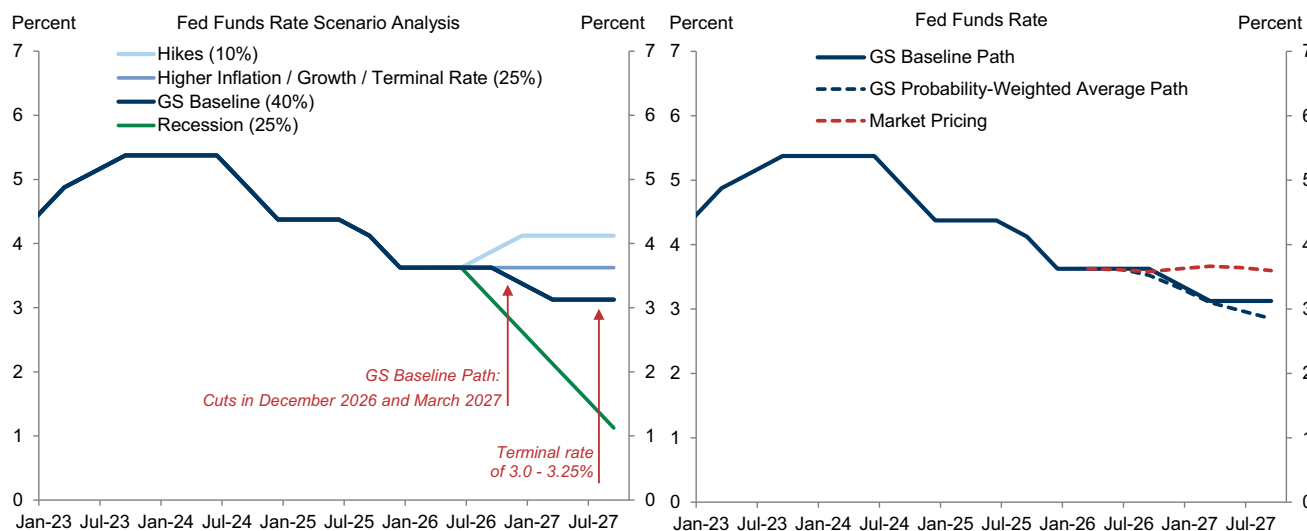
Note: Dashed lines indicate GS forecasts.

Source: Haver Analytics, Goldman Sachs Global Investment Research

7. Upside PCE inflation surprises, better labor market data, and growing hawkishness within the FOMC have led us to delay our forecast for two more 25bp normalization cuts

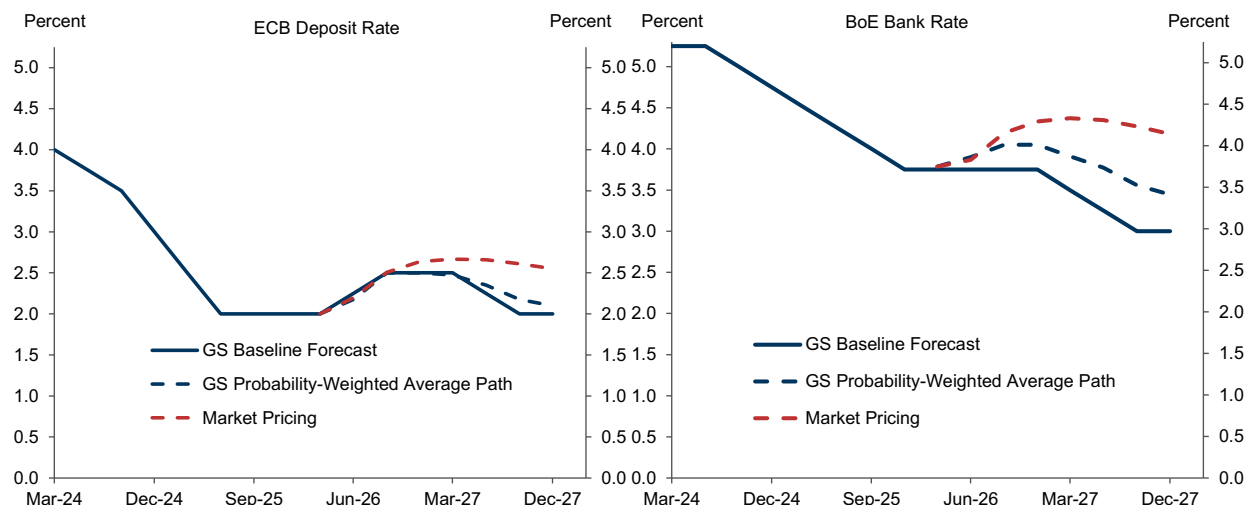
to December 2026 and March 2027. However, our forecast remains below market pricing, especially in probability-weighted terms. One important question will be the Fed leadership transition. We doubt the committee would endorse Kevin Warsh's earlier call for cuts in anticipation of AI-driven disinflation. But his more recent argument that trimmed-mean PCE—which is running 0.8pp below core PCE—is a better measure because it is less sensitive to one-time price shocks seems more promising. After five years of above-target inflation, most sitting Fed officials are reluctant to make this argument publicly, lest they be accused of cherry-picking. But many probably agree with it tacitly and might be persuaded to do so more openly alongside a new Chair.

Exhibit 7: Fed Cuts on a More Delayed Schedule



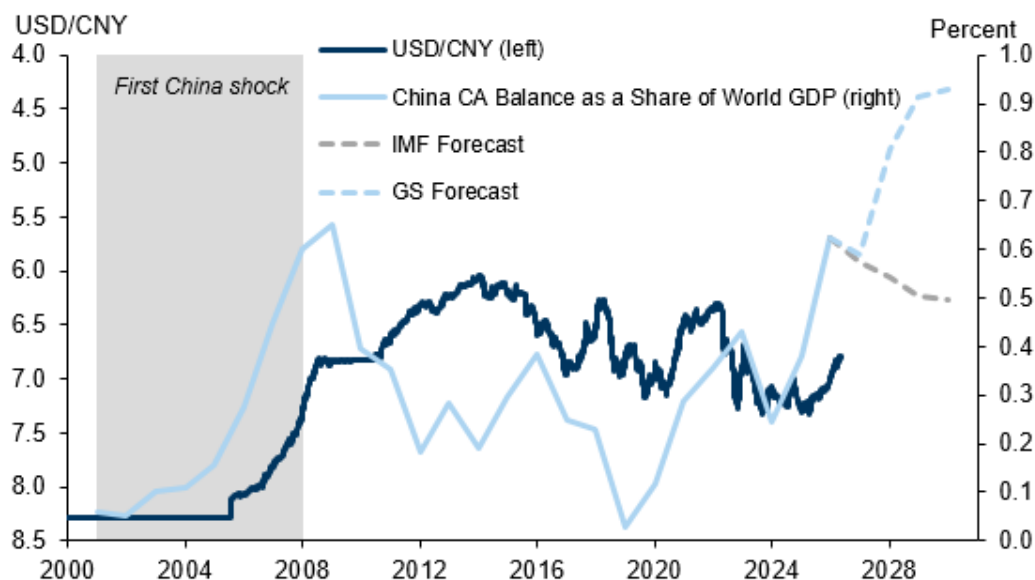
Source: Bloomberg, Goldman Sachs Global Investment Research

8. We still expect the ECB to deliver two 25bp hikes soon—probably in June and September—because of the low 2% starting point for the deposit rate and the historical tendency to (over)react hawkishly to oil-driven headline inflation shocks. However, the risk is on the side of no hikes because core inflation remains very well-behaved so far and ECB officials as a group are more open-minded about core vs. headline than in the pre-Draghi days. In any case, we think that any near-term hikes would probably be unwound in 2027. Across the Channel, we don't see hikes from the BoE in our baseline—in part because Bank Rate at 3.75% is clearly in restrictive territory—but they do constitute a meaningful hawkish risk.

Exhibit 8: We See the ECB Hiking and the BoE on Hold, But Our Views Are Dovish Relative to Market Pricing in Both Cases

Source: Bloomberg, Goldman Sachs Global Investment Research

9. In the run-up to this week's Trump-Xi summit, China's economy continues to benefit from strength in the export-oriented goods sector. This kept Q1 GDP growth at 5.0% year-on-year and our 2026 forecast at 4.7%, but the economy remains very unbalanced. While higher prices for imported energy should temporarily halt the increase in the current account surplus this year, we expect the combination of strong production and sluggish demand to push it further into unprecedented territory as a share of global GDP in subsequent years. The clearest market implication is that CNY is undervalued, especially against the US dollar where our FX strategists estimate a gap of more than 20% and have doubled down on their forecast of significant appreciation, with a new target of 6.50 in 12 months.

Exhibit 9: China's Record Current Account Surplus Should Boost the CNY

Source: Goldman Sachs FICC and Equities, Haver Analytics, IMF, Goldman Sachs Global Investment Research

10. If we wrote down the most frequent topics in discussions with market participants, the vast majority would be negative—an unprecedented oil supply shock, large fiscal

deficits, questions about Fed independence, population shrinkage, cyber/AI security risks, US-China rivalry, and the biggest war in Europe since WW2, amongst others. Moreover, equities are far from cheap. So why have they performed so well? The likely answer is the strength of the Q1 earnings season combined with the improving long-term outlook for productivity and thus profits on the back of the AI revolution. The latter is especially important in the US market, where our strategists estimate that earnings and dividends a decade or more in the future represent roughly 75% of fundamental value. That said, a market where the baseline is positive but the risks are so asymmetrically negative argues for combining long risk position with an exposure to equity volatility or significant hedges (our cross-asset strategists recommend equity puts struck deeply out of the money).

Jan Hatzius

Disclosure Appendix

Reg AC

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AI ADOPTION TRACKER

April 2026: Adoption Rises to 19.8%

- We update our AI adoption tracker through April 2026.
- **AI-related investment growth** remains strong, particularly for semiconductors, where equity analysts expect global revenue growth of 50% from current levels by the end of 2026. AI-related investment in the US national accounts also continues to rise and now stands at \$360bn (1.1% of GDP) above its 2022 level. US investment in AI-related hardware and software continued to rise in today's 2025Q4 GDP release. AI-related hardware shipments from Taiwan to the world rebounded in March and stood at \$61.2bn.
- **AI adoption by firms** rose by 0.9pp to 19.8% among US establishments (with adoption expected to rise to 23.0% in the next 6 months) according to the Census Bureau's Business Trends and Outlook Survey (BTOS). Information services, professional services, education services, and finance/insurance firms continue to lead adoption. The latest BTOS release also included several **new supplemental questions** regarding AI adoption and its impacts on business activity. Sales and marketing, strategy and business development, and R&D are the most common AI use cases among adopters today, and intensity of use of AI tools for these purposes is expected to ramp up sharply in the next 6 months. Most firms are using AI only to supplement a small number of employee work tasks and AI adoption has had a minimal impact on total business employment in most sectors. Among non-adopters, AI incompatibility with the business and a lack of knowledge on AI capabilities were the most commonly cited barriers to adoption.
- **AI's labor market impact** remains narrow. Employment in specific industries like marketing, graphic design, customer service, and especially tech where AI-use cases have been established is contracting by around 11k per month. However, this is offset by construction job growth in data center-exposed categories, which is up 212k since 2022 (relative to broader construction employment trends). The tech share of total employment continues to decline relative to the pre-2022 trend. That said, the correlation between AI adoption and other labor market slack measures remains limited, and only 15.3k employees were affected by corporate layoffs attributed to AI in March (though several other high-profile tech companies have announced layoff plans in the coming months over the last week).
- We continue to observe large **impacts on labor productivity** in the limited areas

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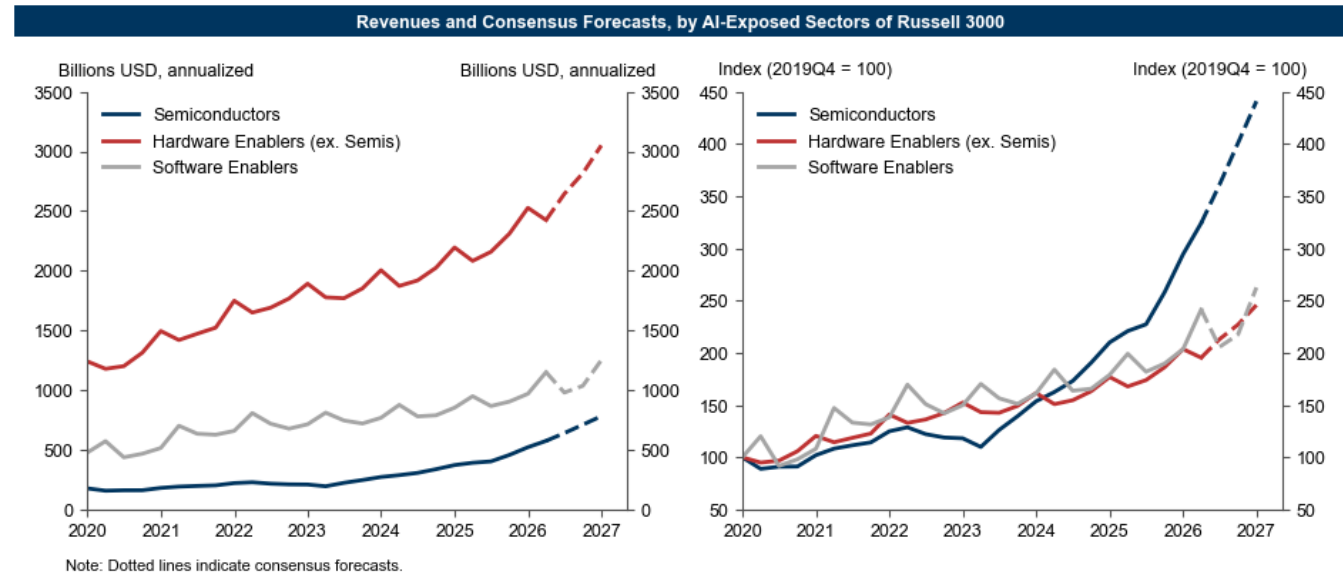
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where generative AI has been deployed. Academic studies imply a 23% average uplift to productivity, while company anecdotes imply slightly larger efficiency gains of around 33%. Industries with higher AI adoption rates are now showing a slight acceleration in productivity growth over the past year in official US data.

April 2026: Adoption Rises to 19.8%

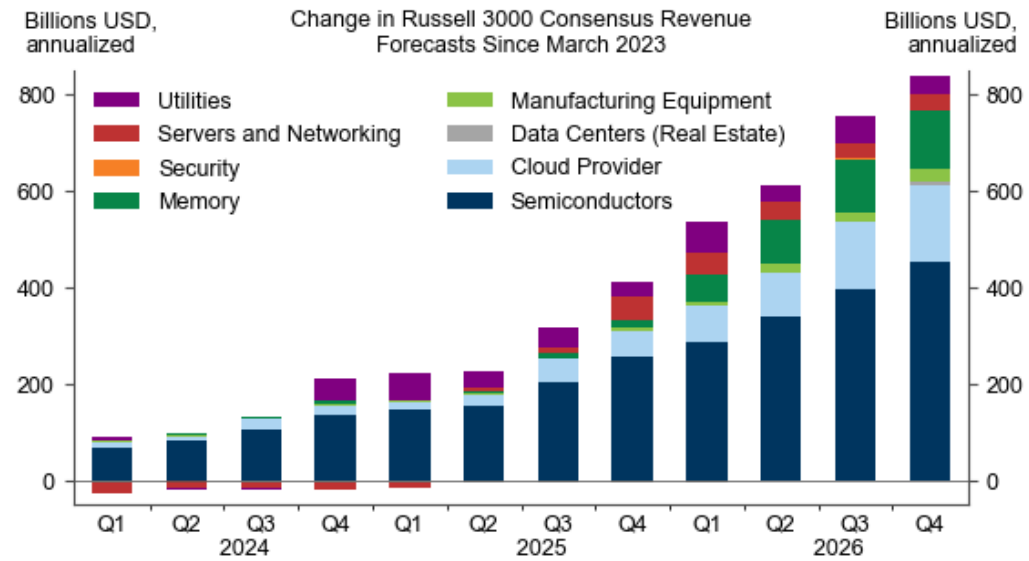
AI Investment

Exhibit 1: Semiconductor Company Revenues Are Expected to Increase by 50% in 2026 on a Q4/Q4 Basis



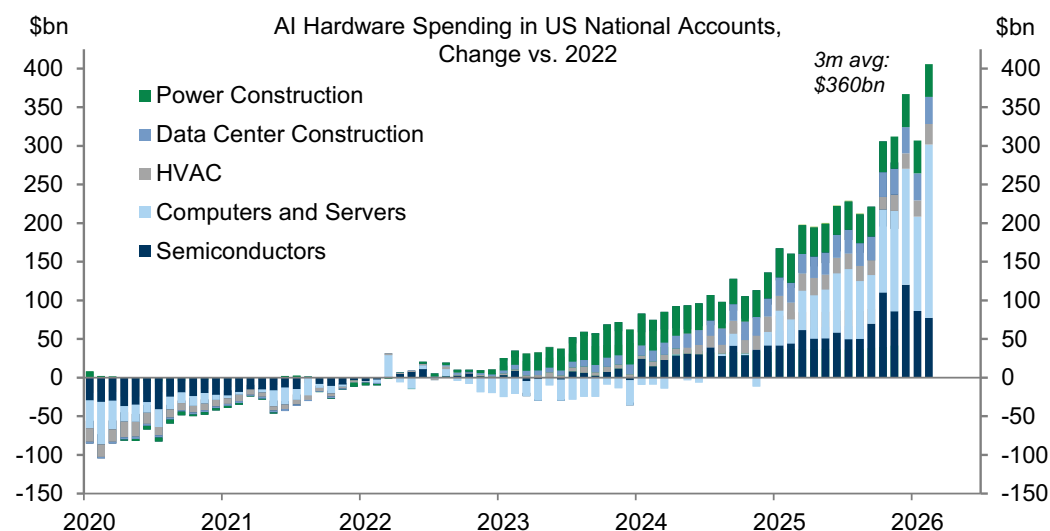
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 2: Revenue Forecast Revisions Imply that AI-Related Revenues Could Rise to Over \$800bn in 2026Q4



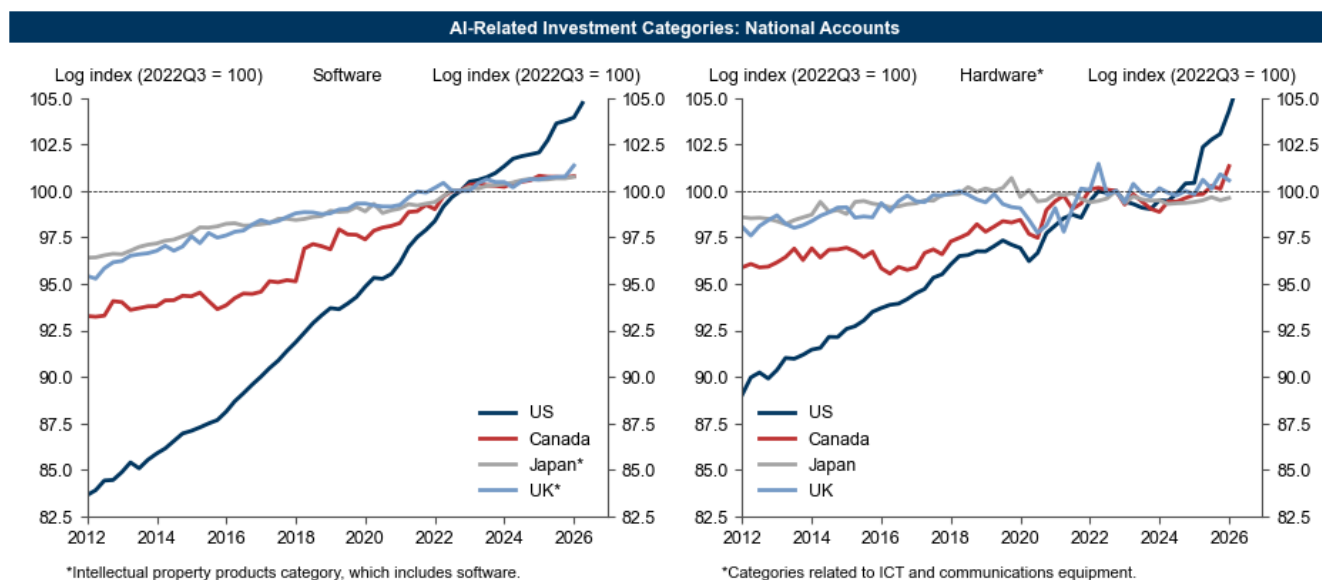
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 3: AI Hardware Investment in the US National Accounts Has Increased by \$360bn Since 2022 on a Three-Month Annualized Basis



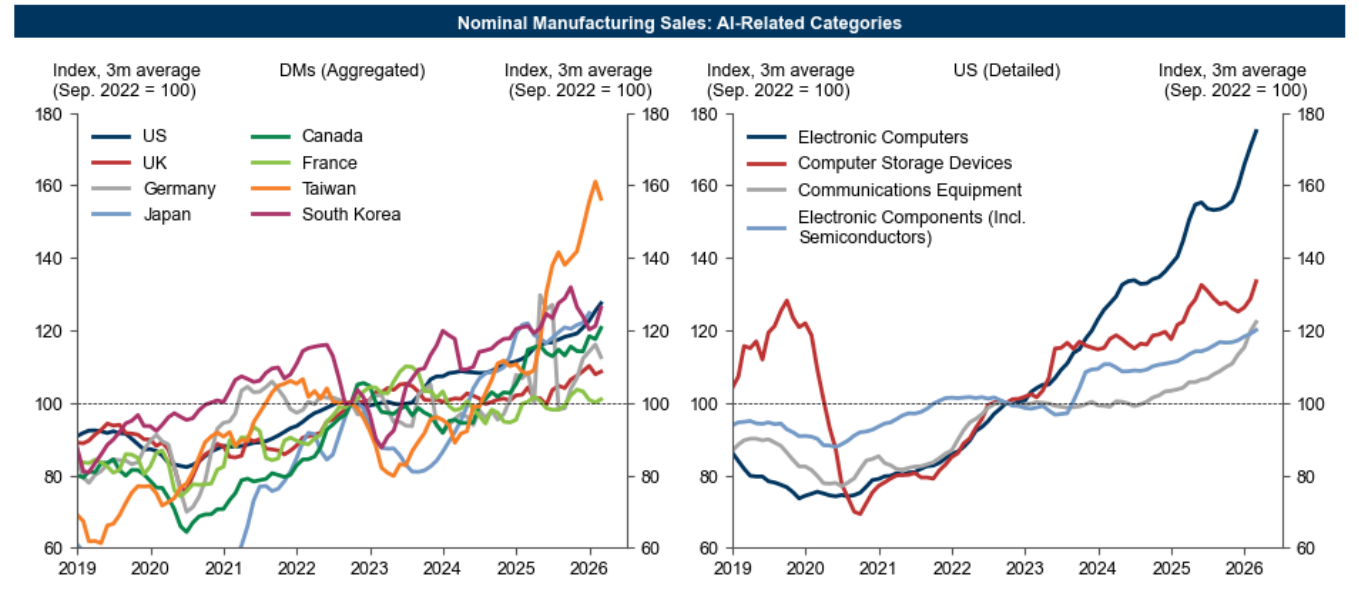
Source: Census Bureau, Bureau of Economic Analysis, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 4: US Investment in AI-Related Hardware and Software Continued to Rise in 2025Q4



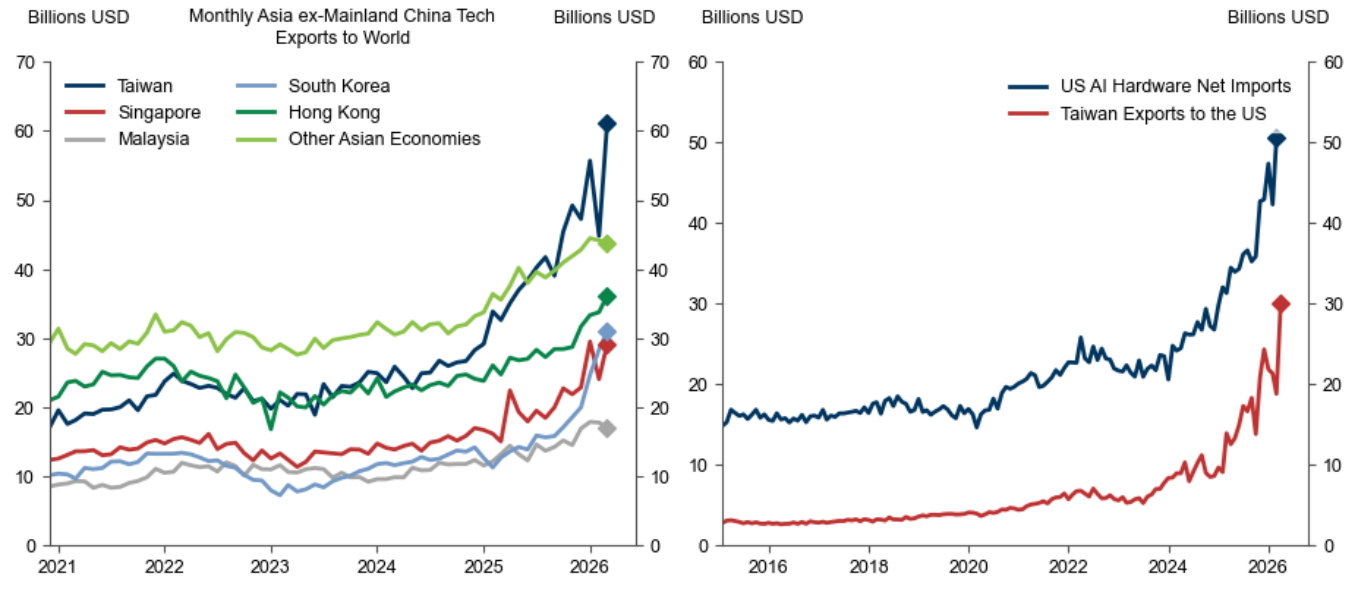
Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 5: AI-Related Manufacturing Shipments Remain Elevated in Taiwan...



Source: Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 6: ...and Taiwan Total Tech Exports to the World Surged to \$61.2bn in March

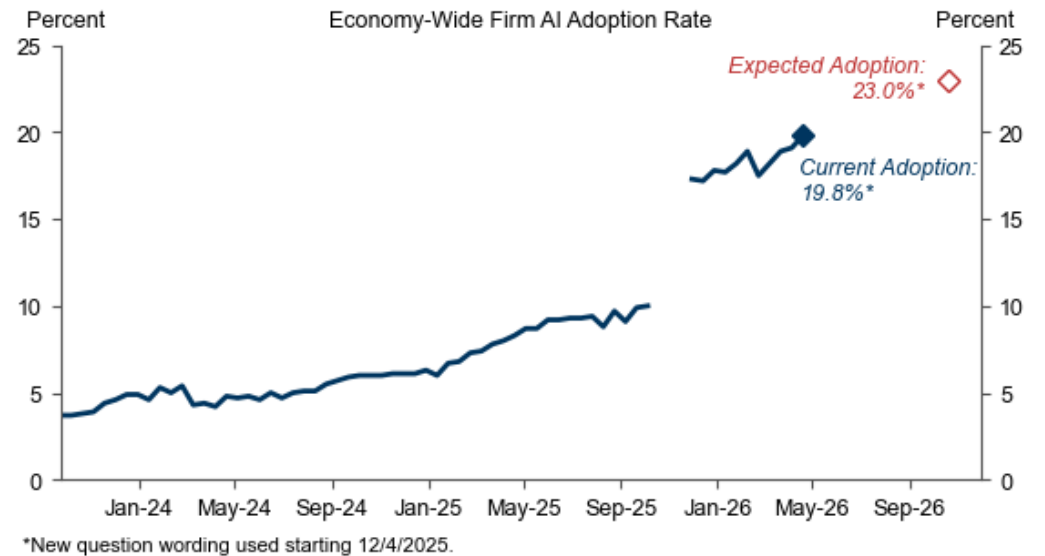


Source: Haver Analytics, Goldman Sachs Global Investment Research

AI Adoption by Firms

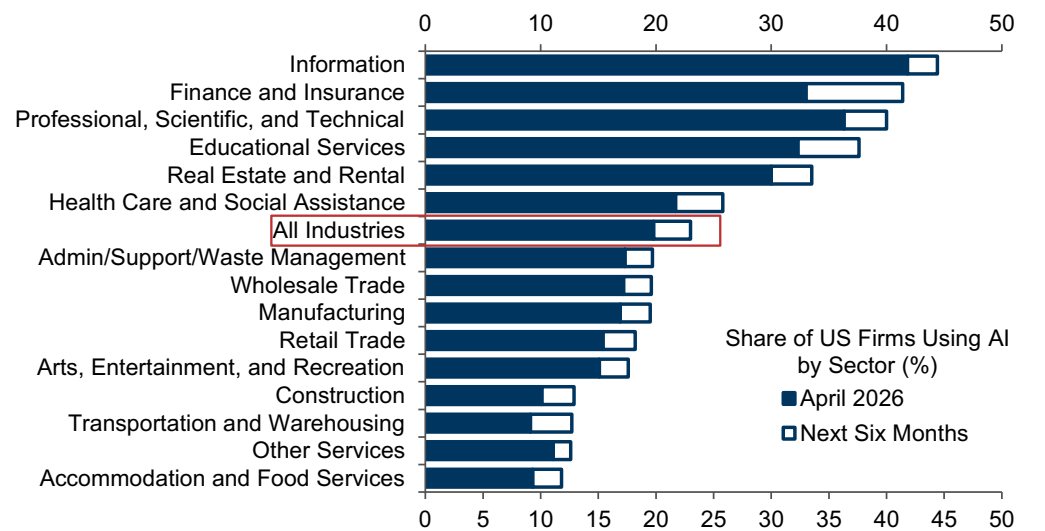
Note: The Census Bureau has rephrased questions from asking about AI use in “producing goods and services” to asking about AI use in “any business functions” starting from the December 4, 2025 release. There was a level shift observed in conjunction with the new question wording, and we will be tracking adoption rates under the new methodology going forward.

Exhibit 7: 19.8% of Firms Are Using Artificial Intelligence in Their Regular Business Functions (+0.9pp From March); 23.0% Expect to Use AI In the Next 6 Months



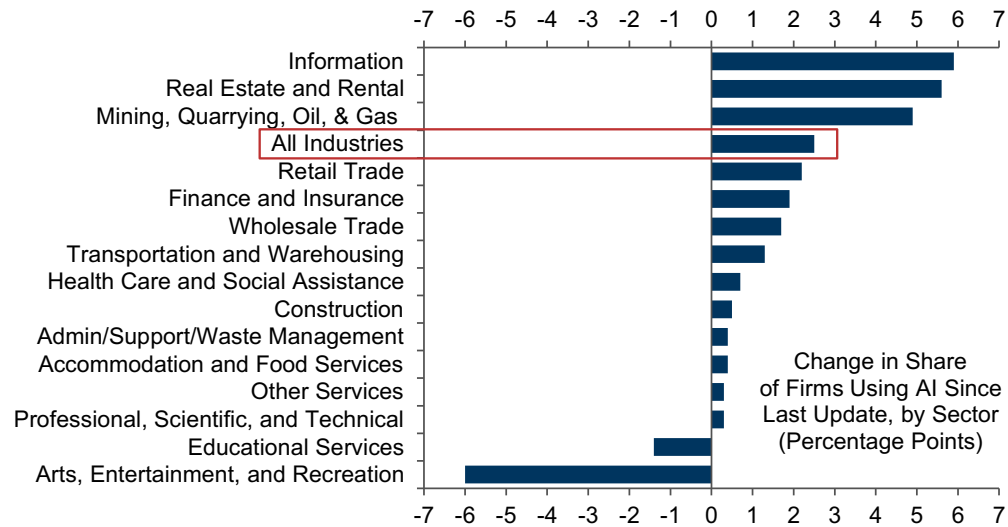
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 8: Information, Professional Services, Education, and Finance and Insurance Establishments Continue to Lead Adoption



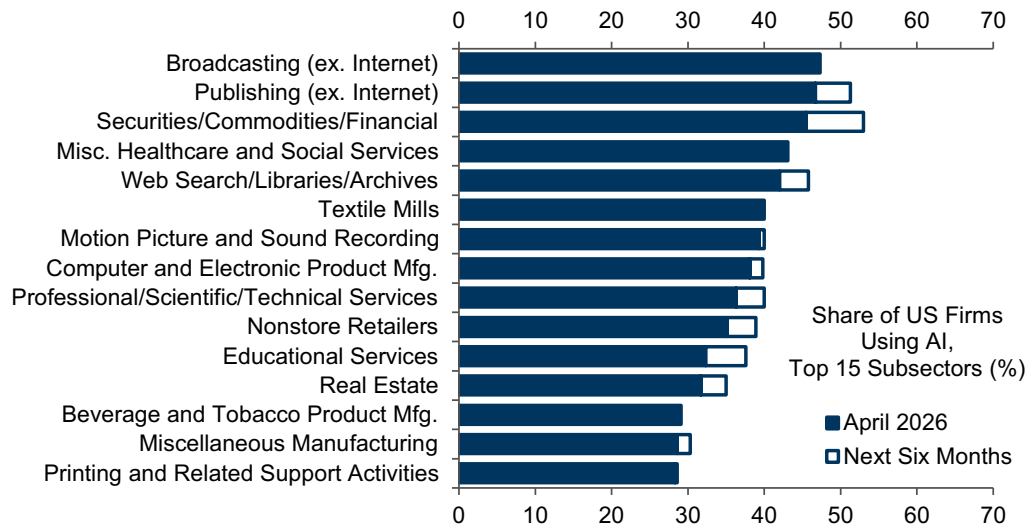
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 9: Information and Real Estate Establishments Had the Largest Gains in Adoption Since Our Last Update



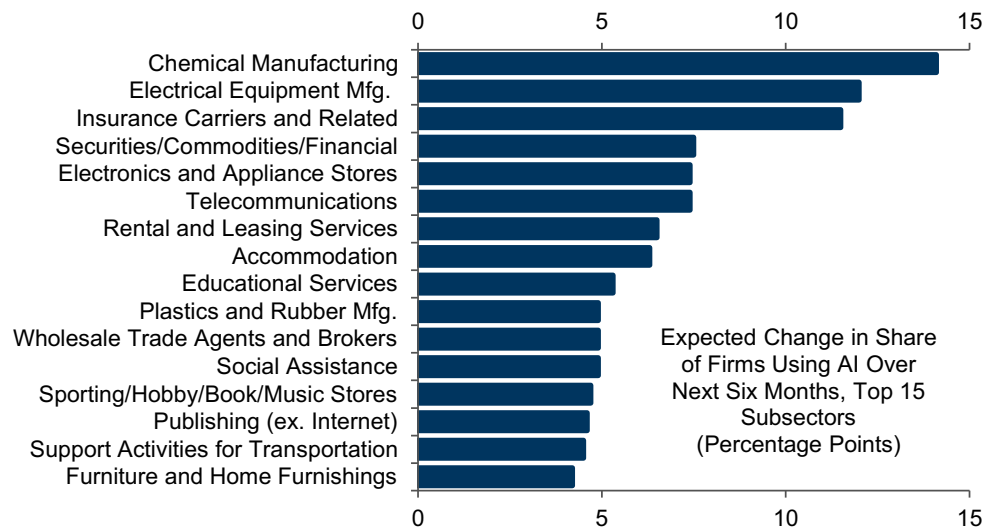
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 10: Among Subsectors, Current Adoption Remains the Highest in Broadcasting, Publishing, and Securities



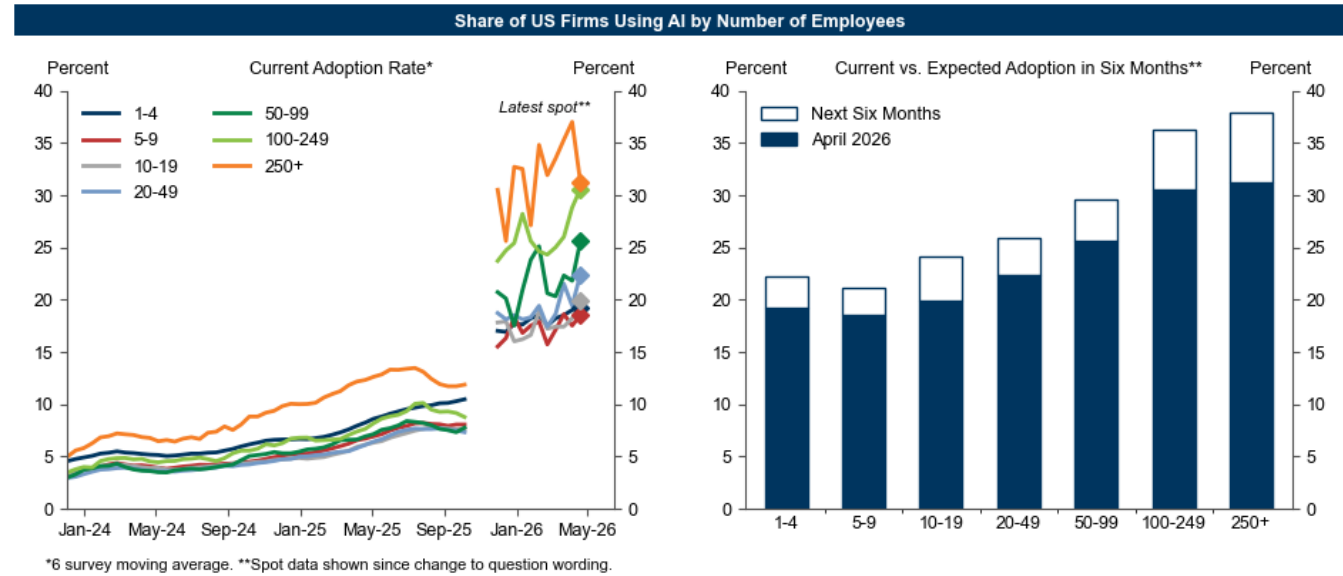
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 11: Chemical and Electrical Equipment Companies Report the Largest Expected Adoption Increases in the Next 6 Months



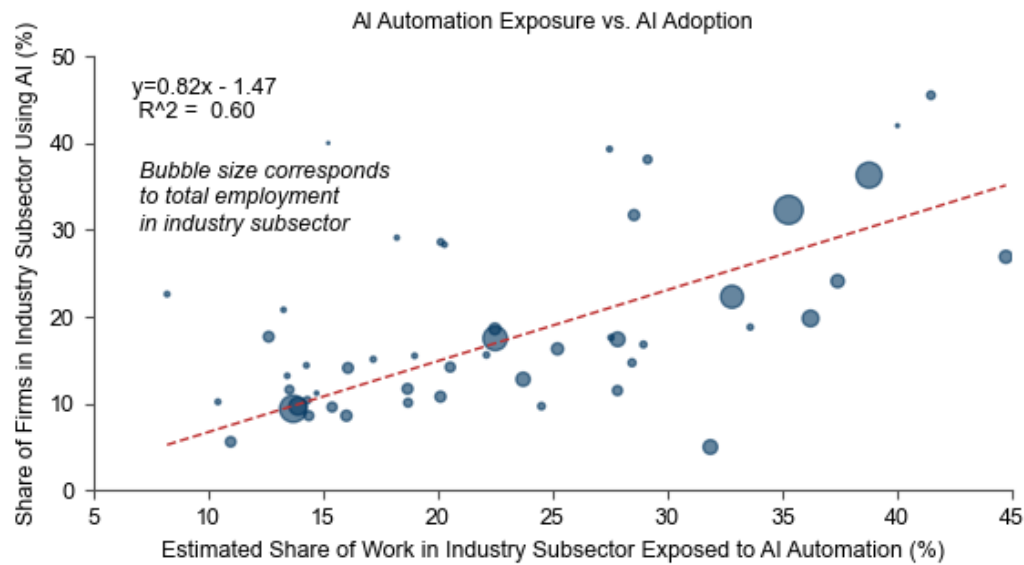
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 12: Current Adoption Among Mid-Sized Firms With 100-249 Employees Rose to 30.6%



Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 13: AI Adoption Remains Strongly Correlated With Our Subsector AI Automation Exposure Scores



Share of work exposed to AI automation is a GS estimate based on the methodology in “Global Economics Analyst: The Potentially Large Effects of Artificial Intelligence on Economic Growth” (March 26, 2023).

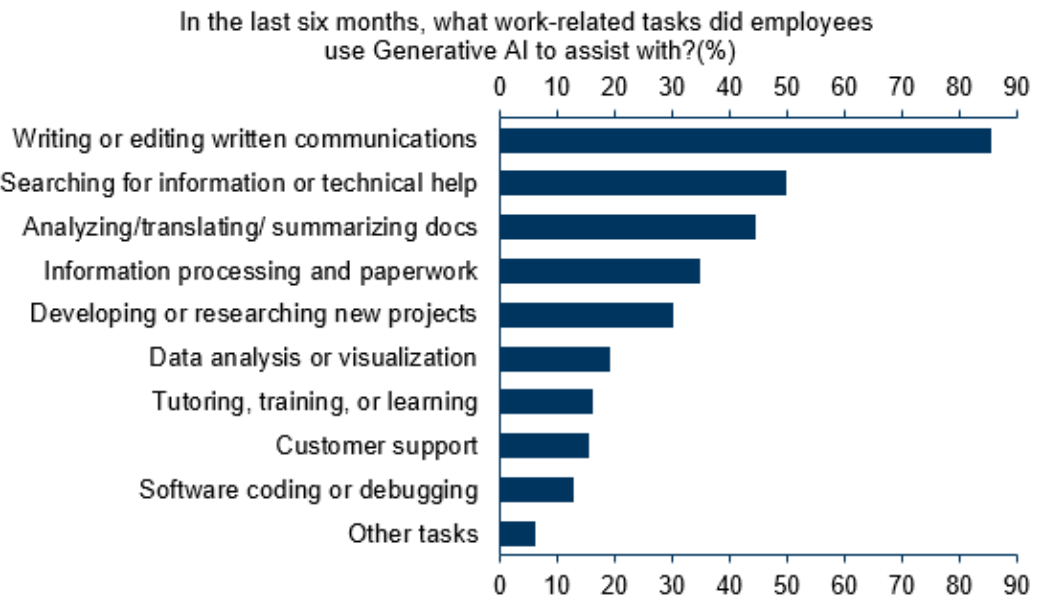
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 14: Sales and Marketing, Strategy and Business Development, and R&D Are the Most Common AI Use Cases...



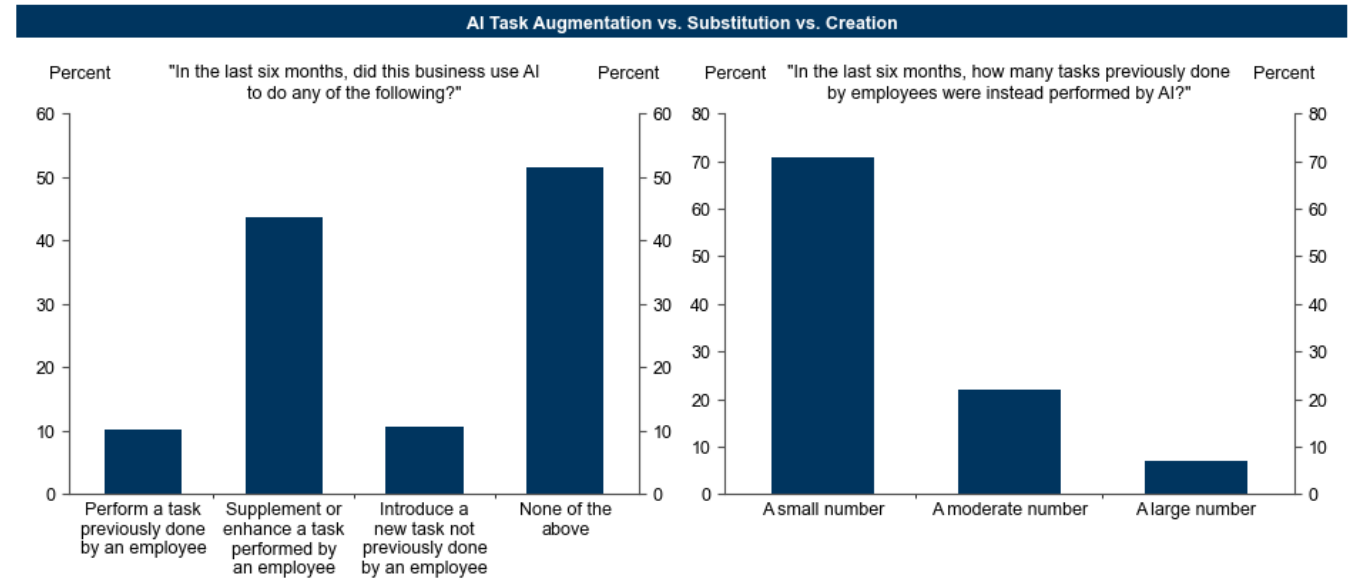
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 15: ...and Writing or Editing Written Communications Was the Most Commonly Cited Work Task that Generative AI Assisted With



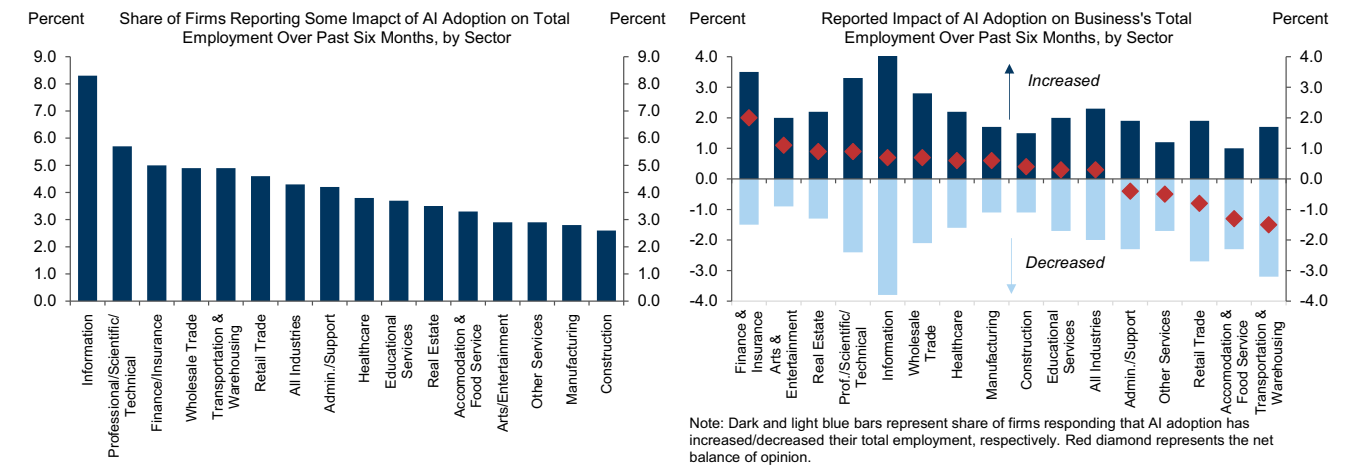
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 16: Most Firms Are Using AI Only to Supplement a Small Number of Employee Work Tasks



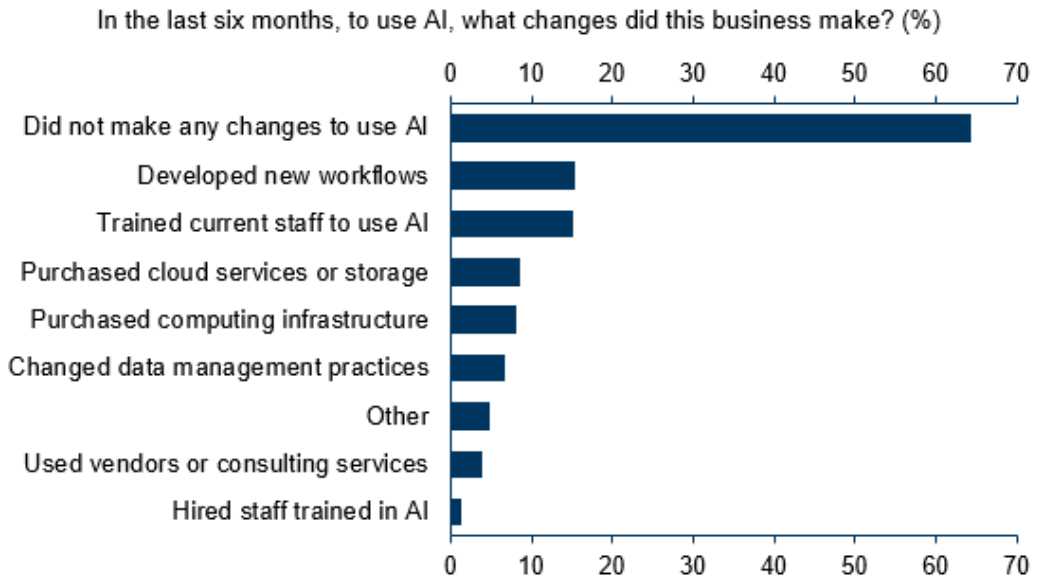
Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 17: Information Services, Professional Services, and Finance and Insurance Firms Are Most Likely to Report that AI has Impacted Headcount in Some Way, but the Net Impact on Employment for the Overall Economy Is Small (and Slightly Positive)



Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 18: Most Firms Did Not Make Significant Organizational or Infrastructure Changes to Adopt AI



Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 19: Inapplicability Was the Most Commonly Cited Barrier to AI Adoption Among Non-Adopters



Source: Census Bureau, Goldman Sachs Global Investment Research

Exhibit 20: A Majority of Surveyed Employees Report Positive Productivity Impacts from Using AI at Work

Source	Date	Current Adoption / Impacts	Future Adoption / Impacts
Gallup	April 2026	Half of all surveyed adults reported using AI in their jobs and 27% of employees in AI-adopting organizations say that their workplace has changed in disruptive ways due to AI in the past year. 65% of adopters said that AI has improved their productivity and efficiency, regardless of how often they personally use AI at work.	Eighteen percent of all U.S. employees say it is very or somewhat likely their job will be eliminated within the next five years due to AI or automation. Among employees working in organizations that have adopted AI, that share rises to 23%. However, employees in AI-adopting organizations were more likely to report both current expansions and reductions in headcount.
NVIDIA	March 2026	64% of respondents said their organizations are actively using AI in their operations. 28% said they're still in the assessment phase, while 8% said they're not using AI and have no plans to. Regional adoption among respondents was 70% in North America, 65% in EMEA, and 63% in APAC. Larger companies (with over 1000 employees) demonstrated broader adoption, deployed more use cases, and reported greater ROI. More than half of respondents (53%) said improved employee productivity was one of the biggest impacts AI had on business operations.	86% of respondents said their AI budgets would increase this year and nearly 40% said budgets would increase by 10% or more. Respondents reported that workflow and operations and retaining the right expertise to scale AI solutions were the key challenges to adoption.
Thomson Reuters	February 2026	40% of respondents said their organizations were using generative AI, up from 22% a year ago. Publicly available tools such as ChatGPT are still the most commonly used platform. In contrast, only 15% of respondents said their organization use agentic AI tools.	53% of respondents said their organizations are either in the planning or consideration phase for agentic AI. 77% of professionals say they expect it to be a central part of their workflow by 2030.
Deloitte	January 2026	In the latest survey, 34% of companies were starting to use AI to deeply transform their businesses, 30% were designing key processes around AI, and the remaining 37% were using AI at the surface level with little or no change to underlying business processes. 84% of companies have not redesignated jobs or work tasks around AI capabilities. 25% of surveyed leaders reported that AI is having a transformative impact on their organizations, more than double from 12% a year ago.	Survey results showed that insufficient employee skills were the largest barrier to integrating AI into existing workflows. Nearly three-quarters of surveyed companies expect to be using agentic AI at least moderately in two years.
Gallup	December 2025	The percentage of US employees who reported using AI at work at least a few times a week increased from 19% in 2025Q2 to 23% in Q3. Unsurprisingly, employees working in knowledge-based jobs (such as tech or professional services) were more likely to use AI than those in frontline positions. Most common use cases include using AI to consolidate information or data, idea generation, and automating basic tasks. Users most often use AI chatbots and virtual assistants.	Gallup research has shown that employee AI adoption is strongly associated with having greater managerial support for AI and strategic integration of AI in their role, indicating that a strategic people-first approach is needed for effective AI adoption.
McKinsey	November 2025	88% of surveyed organizations are regularly using AI in at least one business function, up from 78% last year. However, most surveyed organizations are still in the experimentation / piloting phase of AI and nearly two-thirds have not begun scaling AI across the enterprise. 39% reported experimenting with AI agents specifically and the use of AI agents is most commonly reported among TMT and healthcare businesses. Respondents reported the greatest revenue benefits from AI in marketing and sales, strategy and corporate finance, and product development. A plurality of respondents had little to no change in employee headcount due to AI use over the past year.	32% of respondents expect an overall workforce reduction of 3% or more due to AI use and larger organizations are more likely to expect workforce reductions. Inaccuracy is the AI-related risk that was most commonly experienced and the top priority that organizations are working to mitigate going forward.
Wharton Human-AI Research and GBK Collective	October 2025	Usage of generative AI among enterprise leaders has risen sharply this year and stands at 46% (+17pp yoy). The most common use cases include data analysis, document summarization, and document editing / writing. Nearly three-quarters already see positive ROI on generative AI and four in five expect positive returns within two to three years. Larger enterprises with over \$2B in annual revenue were more likely to report that it is "too early" to measure productivity outcomes of their AI initiatives as they navigate integration complexity.	88% of surveyed business leaders expected an increase in AI spend at their organization in the next 12 months (+16pp yoy). 11% of these leaders are funding AI by cutting spending in other areas, with reallocation most commonly coming from legacy IT and HR programs.
Gartner	June 2025	45% of surveyed firms with high AI maturity reported that their AI initiatives remained in production for 3+ years to ensure sustained impact and value, compared to 20% in low-maturity organizations. 91% of leaders from high-maturity organizations have appointed dedicated AI leaders.	Security threats, finding the right use cases, and data availability / quality were among the top barriers for AI implementation reported.
University of Maryland Center for Global Business	June 2025	58% of the business executives and middle managers surveyed reported that their firms had incorporated AI into their business operations in some capacity. 39% reported using generative AI, 30% reported using computer vision, and 27% reported using machine learning. A majority of respondents said that customer experience and operations efficiency improvements were at the core of their AI adoption strategies	A majority of respondents declared to strongly or somewhat support AI regulations on transparency and/or algorithmic bias. 20% of respondents were very or extremely concerned about AI-driven job displacement in the near future.
Paypal and Reimagine Main Street	June 2025	Out of nearly 1000 small businesses surveyed, 25% reported having already integrated AI into their daily operations while another 51% reported experimenting with AI (but not incorporating into regular operations). Business automation, financial intelligence, and real-time customer insights were cited as some of the most common strategic applications of AI tools.	Respondents who had not yet fully adopted AI cited security concerns, resource constraints, and uncertain value add as the top barriers faced. Most respondents who had not yet adopted AI were in varying stages of engagement on a trajectory towards adoption.

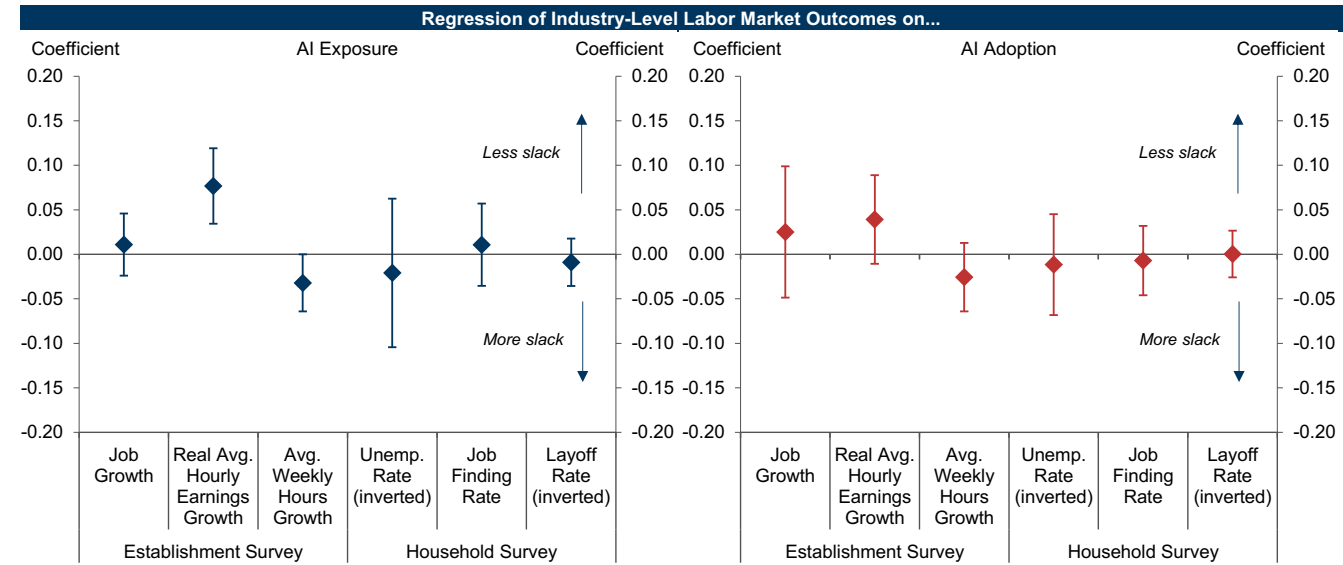
Green shading indicates surveys added since the last quarterly update.

Source: Goldman Sachs Global Investment Research

AI’s Impact on the Labor Market

Labor Outcomes

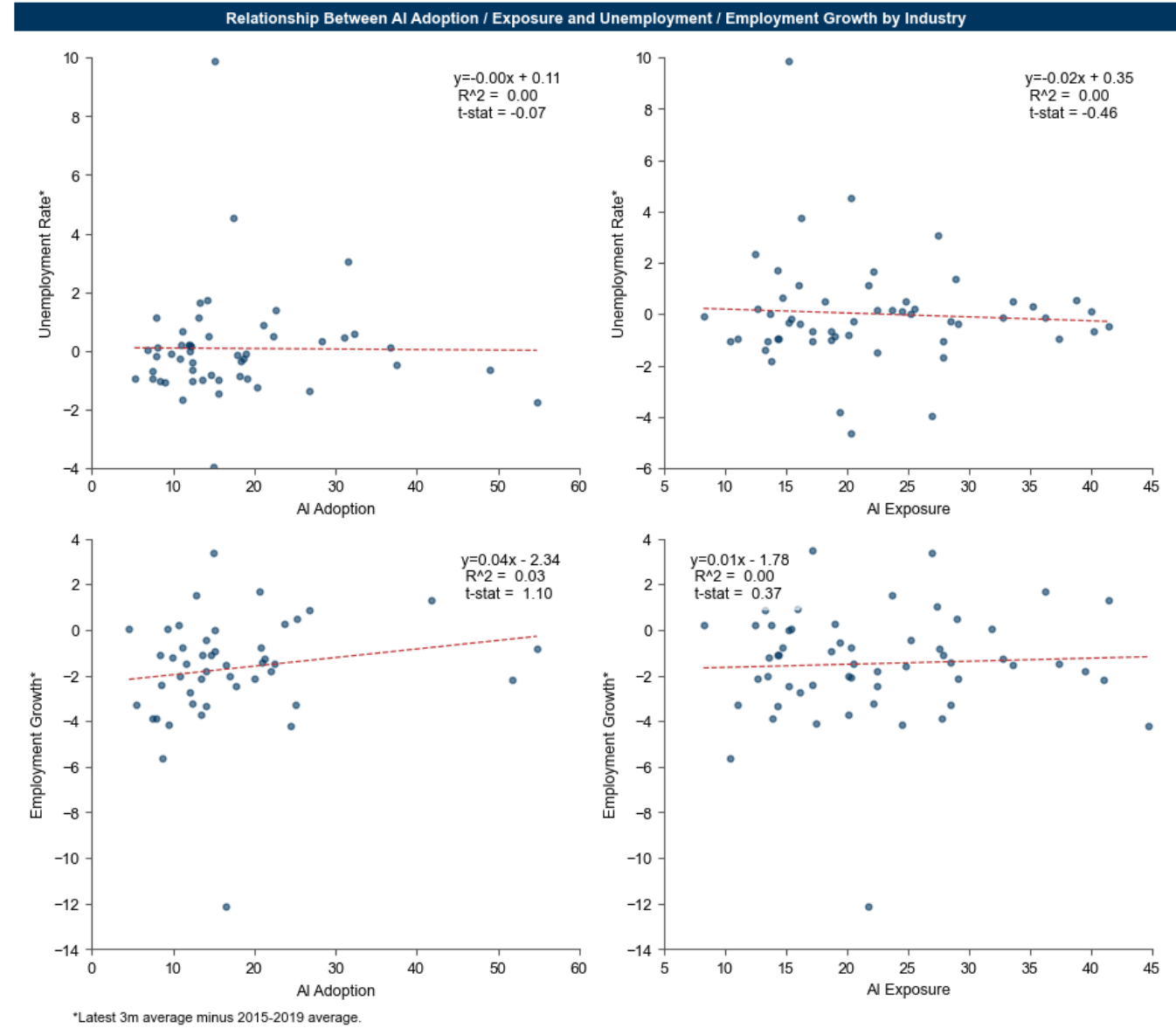
Exhibit 21: The Impact of AI Exposure and Adoption on Broad Labor Market Outcomes Remained Statistically Insignificant Through March



Note: All dependent variables are taken as the deviation from 2015–19 averages. Average coefficient over the past three months is plotted.

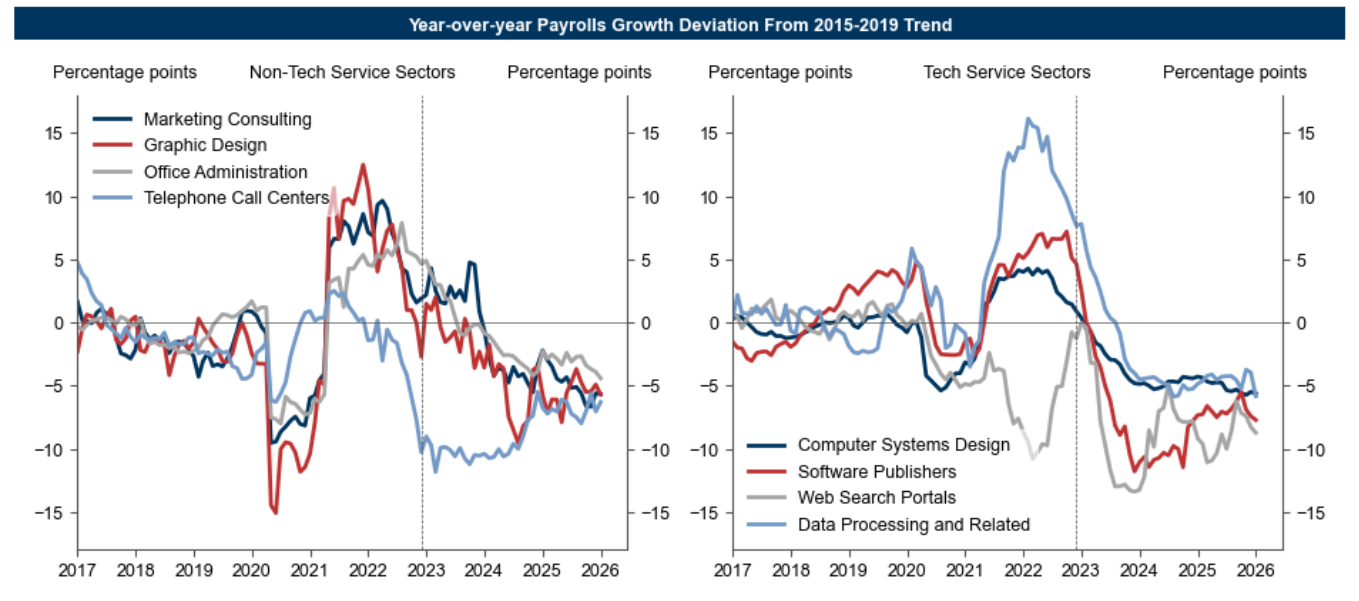
Source: IPUMS, US Bureau of Labor Statistics, Census Bureau, Goldman Sachs Global Investment Research

Exhibit 22: Still No Statistically Significant Relationship Between AI and Unemployment or Employment Growth



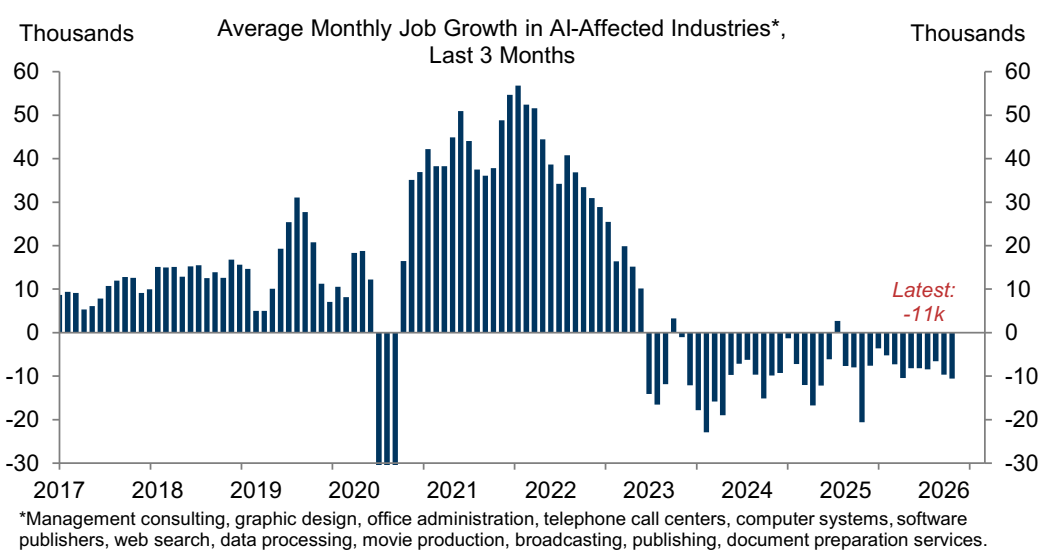
Source: US Bureau of Labor Statistics, IPUMS, Goldman Sachs Global Investment Research

Exhibit 23: Employment Growth Continues to Underperform in Occupations Where Anecdotes Suggest AI Is Having an Impact...



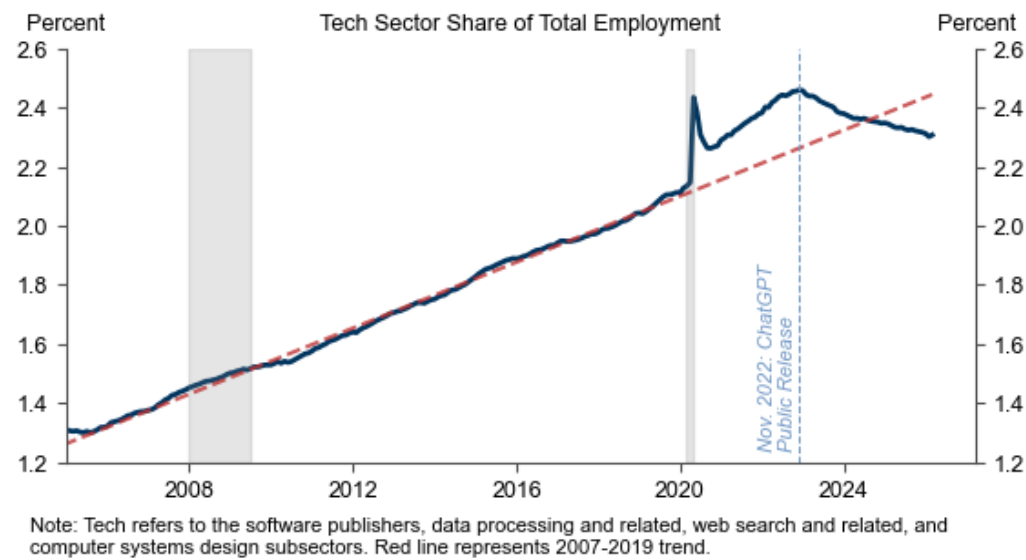
Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 24: ...With a 11k Overall Headwind to Job Growth From Industries Where Use Cases Have Been Established



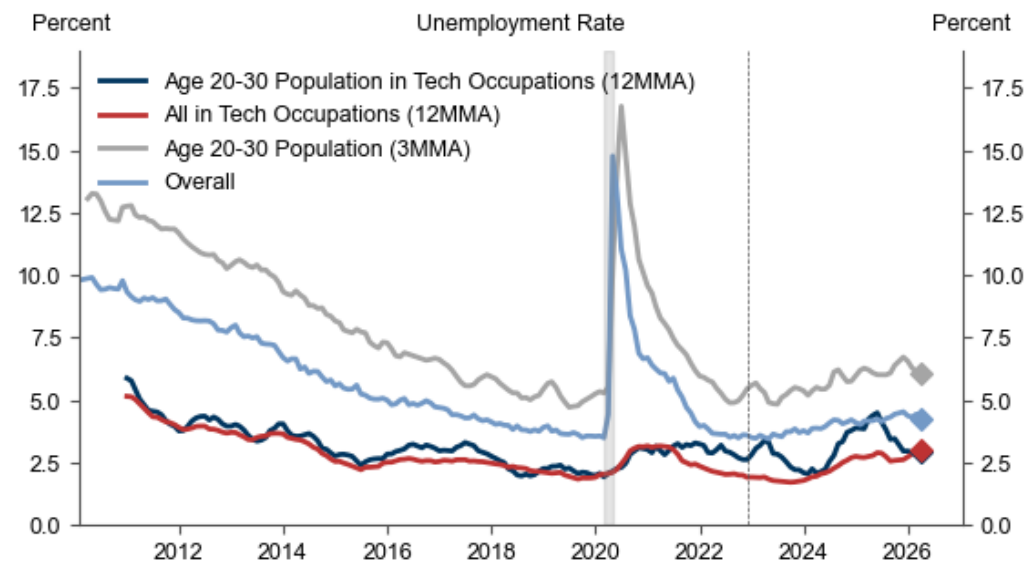
Source: US Bureau of Labor Statistics, Goldman Sachs Global Investment Research

Exhibit 25: Tech’s Employment Share Remains Low Relative to Its Long-run Trend



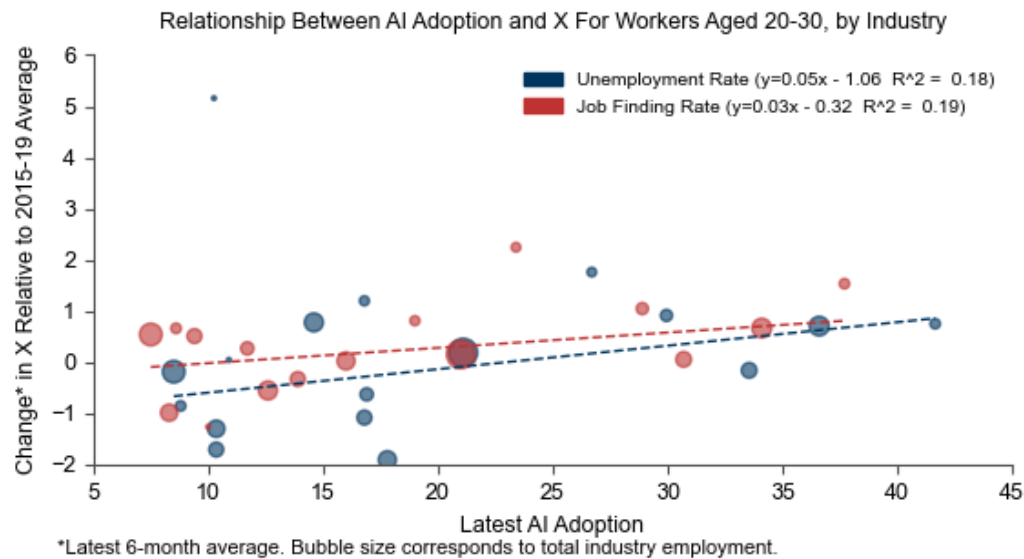
Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 26: Unemployment Among Young Tech Workers Has Fallen Back in Line With Unemployment for All Tech Workers



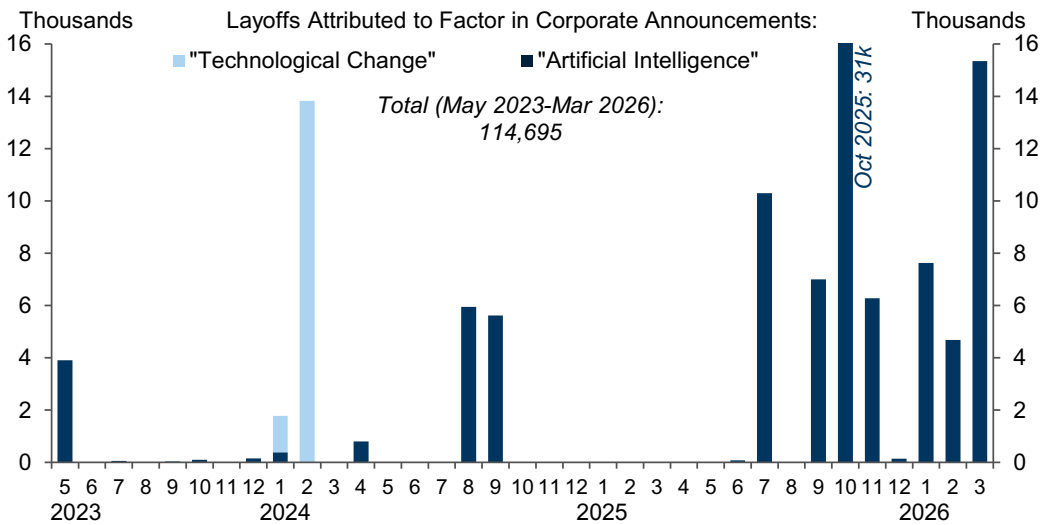
Source: US Bureau of Labor Statistics, IPUMS, Goldman Sachs Global Investment Research

Exhibit 27: We Continue to Observe a Slight Positive Correlation Between AI Adoption and Unemployment Among Younger Workers



Source: IPUMS, US Bureau of Labor Statistics, Census Bureau, Goldman Sachs Global Investment Research

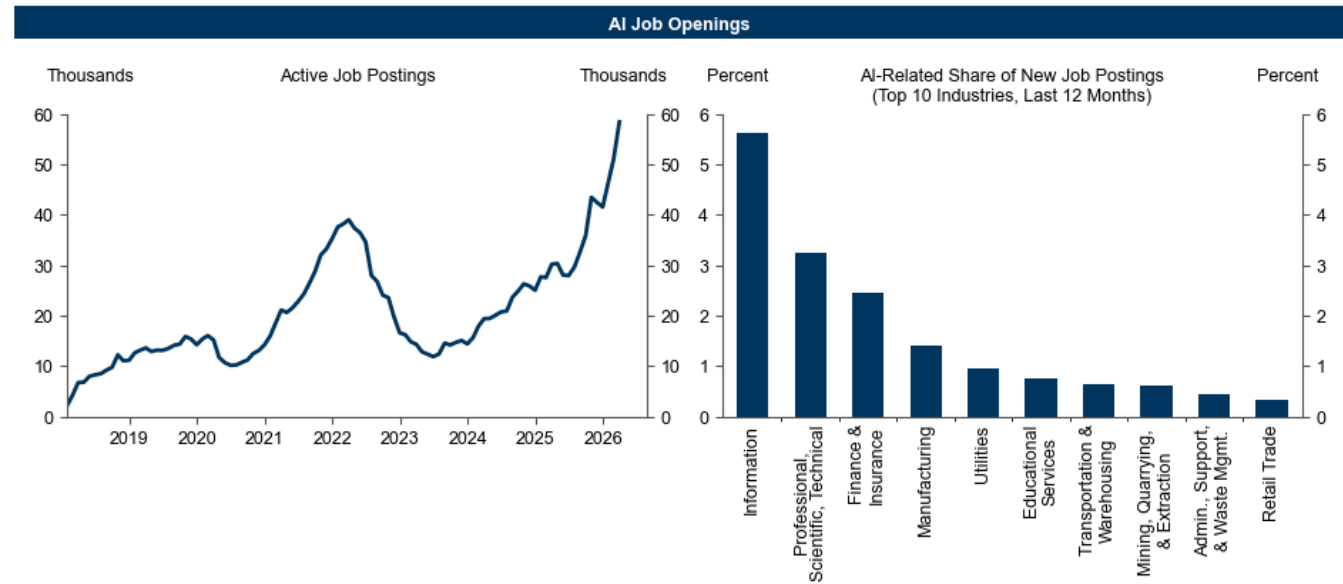
Exhibit 28: AI Was Cited in Layoff Announcements Affecting 15.3k Employees in March



Source: Challenger, Gray & Christmas, Goldman Sachs Global Investment Research

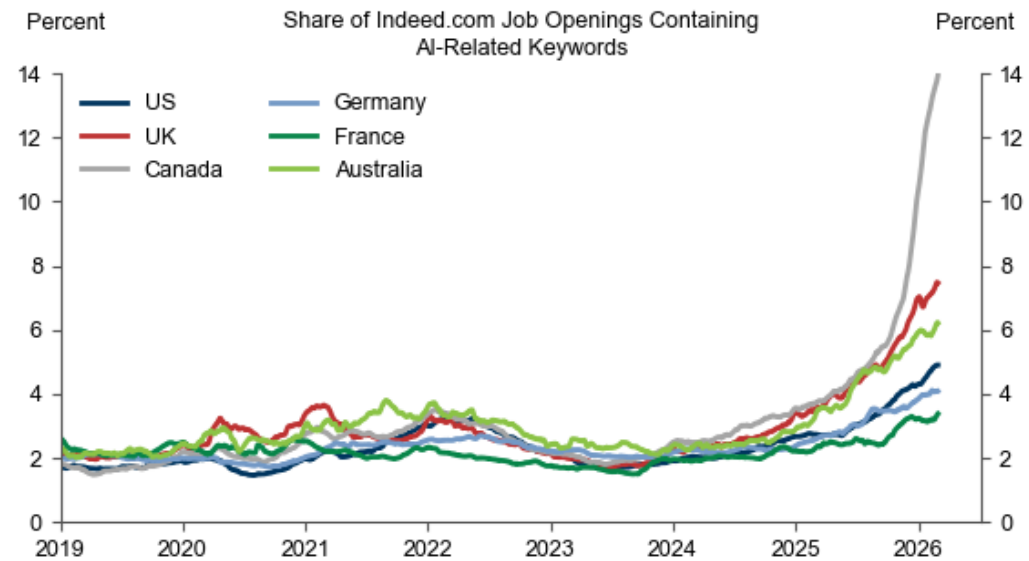
Labor Demand

Exhibit 29: AI Job Postings Remain Skewed Toward Information and Professional Services, as Well as Finance



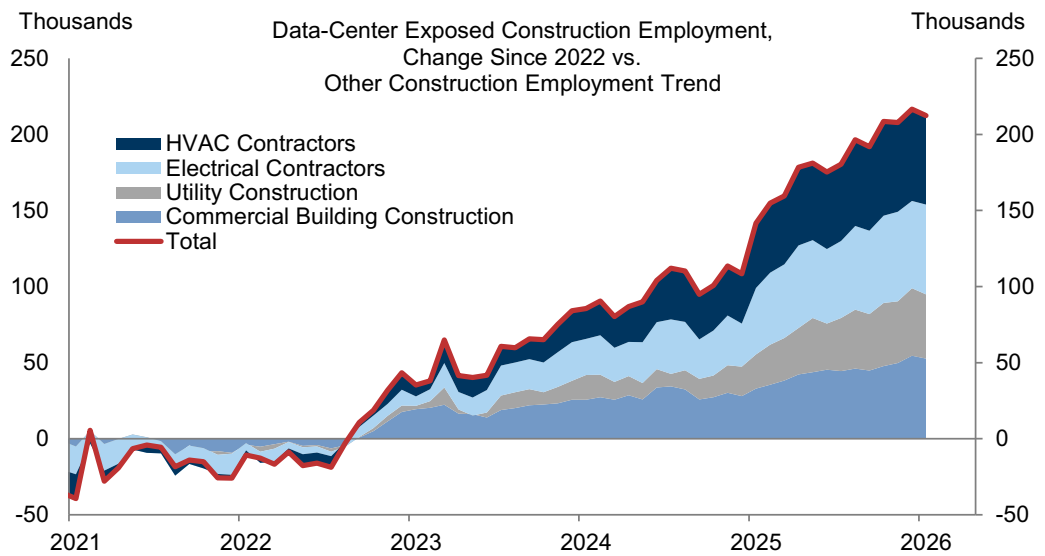
Source: University of Maryland, LinkUp, Goldman Sachs Global Investment Research

Exhibit 30: Share of AI-Related Job Postings Has Risen in Most DMs and Especially in Canada



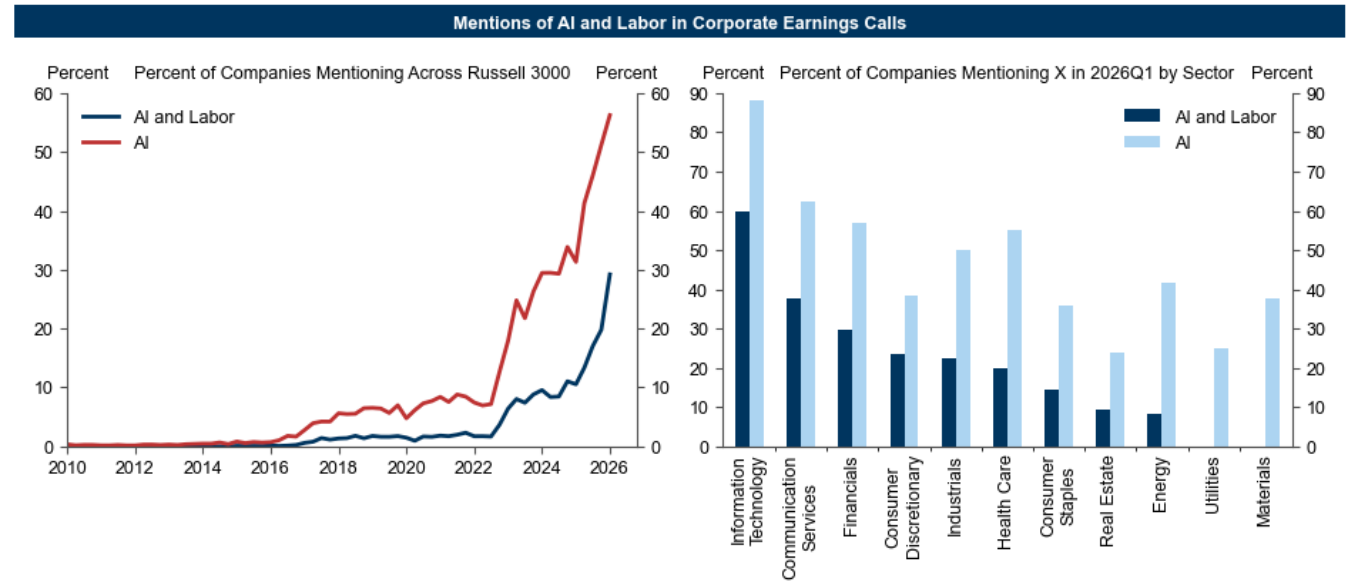
Source: Indeed Hiring Lab, Goldman Sachs Global Investment Research

Exhibit 31: Construction Employment in Data Center-Exposed Categories Has Increased by 212k Since 2022 (Relative to Broader Construction Employment Trends)



Source: US Bureau of Labor Statistics, Haver Analytics, Goldman Sachs Global Investment Research

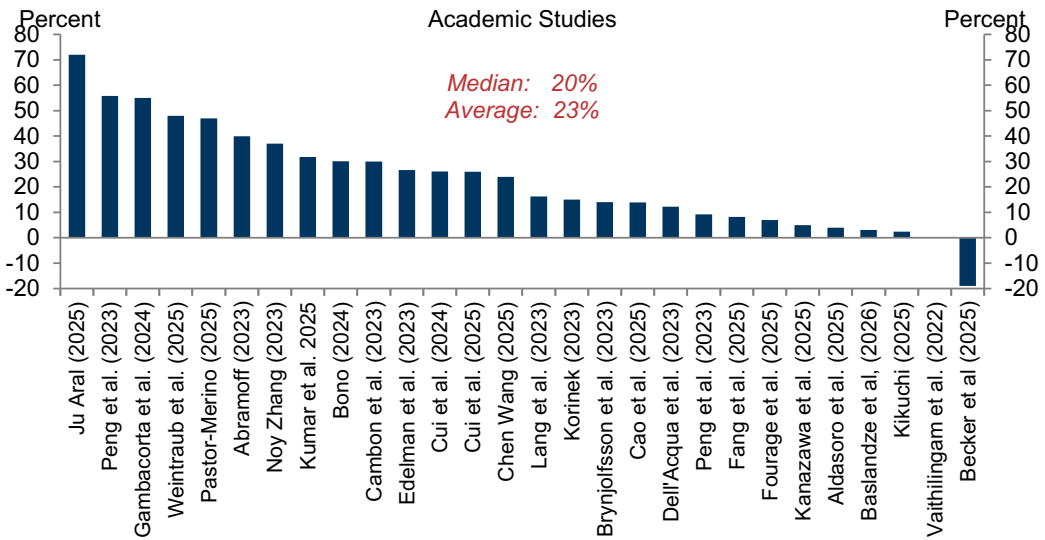
Exhibit 32: Nearly 60% of IT Companies Mentioned AI and Labor-Related Keywords in 2026Q1 Earnings Calls



Source: GS Data Works, Goldman Sachs Global Investment Research

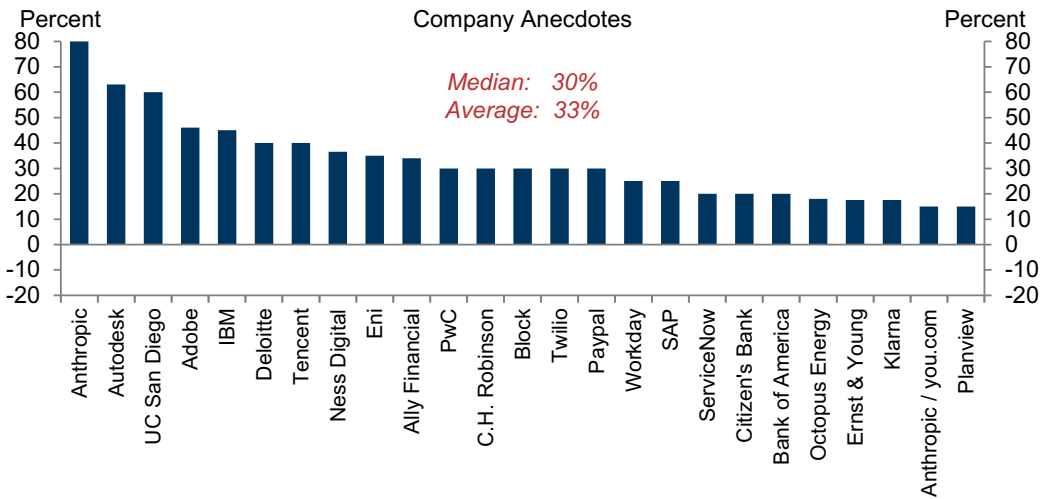
AI’s Impact on Productivity

Exhibit 33: Evidence From Academic Studies Suggests AI Adoption Boosts Labor Productivity by Around 23% on Average...



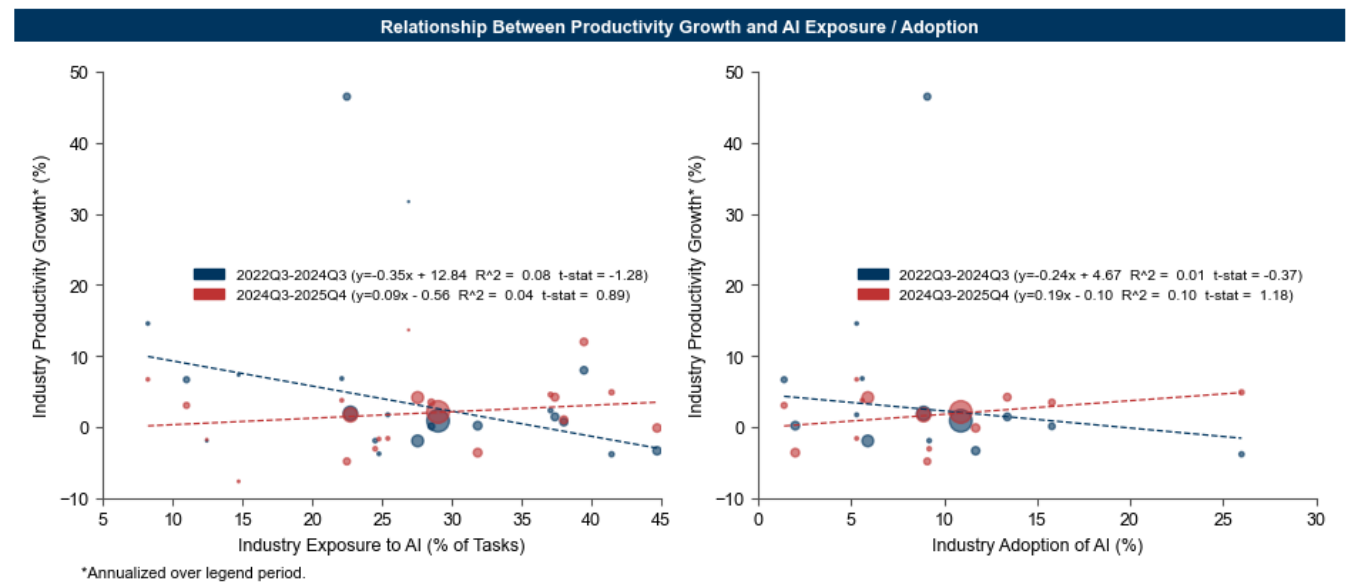
Source: Goldman Sachs Global Investment Research

Exhibit 34: ... And Company Anecdotes Suggest an Average Productivity Boost of 33%



Source: Company data, Data compiled by Goldman Sachs Global Investment Research

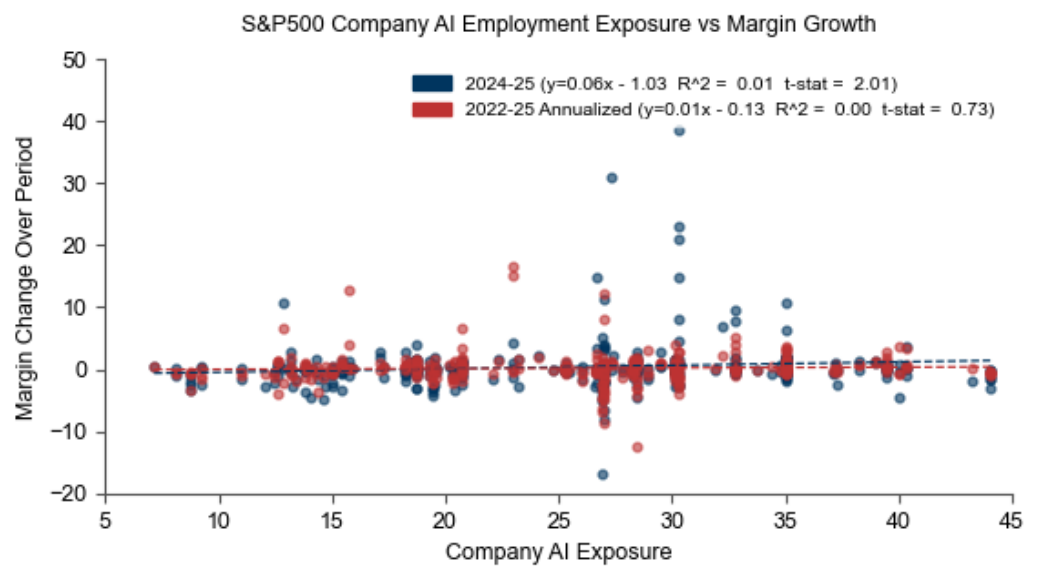
Exhibit 35: Industries with Higher Adoption rates Show Slightly Stronger Productivity Growth Over the Past Five Quarters



Note: Bubble size corresponds to total industry employment.

Source: US Bureau of Labor Statistics, Bureau of Economic Analysis, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 36: No Relationship Between Company Level AI Exposure and Margin Growth



Source: FactSet, Goldman Sachs Global Investment Research

Latest GS Research on Artificial Intelligence

- [Global Credit Trader: A closer look at AI-related debt supply](#), April 30
- [AI Spend vs. Benefit: Marking our views to market two years later](#), April 30
- [Credit Notes: AI-related debt issuance: The public vs. private mix](#), April 21
- [Global Strategy Views: The Technology Value Opportunity](#), April 7
- [US Daily: Breaking Down the Impact of AI on the Labor Market: Substitution vs. Augmentation](#), April 4
- For research from prior quarters, see the [Artificial Intelligence](#) page on our publishing portal.

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Disclosure Appendix

Reg AC

We, Sarah Dong and Joseph Briggs, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

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OIL ANALYST

How Fast Can Gulf Production Recover After Reopening?

- How fast can Gulf crude oil production, which we estimate to be down by 14.5mbd (or 57%) from pre-war levels ([Exhibit 1](#)), recover after the Strait of Hormuz reopens? We find that Gulf production is likely to mostly recover within a few months of reopening assuming 1) no renewed strikes on oil assets and 2) a full and safe reopening of the Strait in coming months. However, we see significant risks that the last leg of the recovery will take significantly longer and may not fully materialize, especially if the Strait were to remain closed for much longer.
- **Transportation and well flow rates are key.** Once the Strait safely re-opens, the key potential constraints on production will likely be the availability of 1) pipeline capacity and empty vessels to destock previously produced oil, and 2) of materials and workers for field workovers, and 3) well flow rates. We estimate that available empty tanker capacity in the Gulf has fallen by around 50% (130mb, #) since the start of the war. We also estimate historical peaks for flows through Hormuz of 23.3mb/d (vs. 20mb/d normal) and for pipeline redirection of 3.5mb/d above normal. Forced curtailments can lead to reservoir complexities in flow rates to varying degrees depending on the reservoir type, requiring intervention and workover jobs prior to opening up wells, to help restore prior production rates as much as possible ([Exhibit 2](#), quotes 1 and 2).
- **The longer the closure, the slower the production ramp.** The longer the Strait is effectively closed, the longer curtailments last, the slower the production recovery is likely to be ([Exhibit 2](#), quotes 2 and 3). This reflects potentially more work on wells, slower procurement of depleted inputs (e.g. drill pipes), and destocking constraining transport capacity for new production for longer.
- **The case for a solid recovery.** We see three reasons why Gulf production is likely to mostly recover within a few months after a full and safe reopening of the Strait:
 - a. The publicly reported evidence on physical damage to oil fields remains limited ([Exhibit 1](#)) in contrast to LNG assets.
 - b. Comments from the Saudi Aramco President and CEO in March suggested that Saudi production is likely to ramp up relatively quickly ([Exhibit 2](#), quote 5).
 - c. The history of supply disruptions ([Exhibit 5](#)) suggests that Saudi Arabia and

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the UAE are likely to deploy available spare capacity ([Exhibit 6](#)) to stabilize markets.

- **Risks to long-run production.** We see three reasons why a full recovery may take several quarters and risks that the recovery may be only partial after a prolonged closure.
 - a. There are large differences across:
 - Gulf fields in terms of reservoir characteristics, infrastructure sophistication, maintenance, and sanctions risks (e.g. Iran). For instance, the share of country production with relatively low reservoir pressure is estimated to be higher in Iran and Iraq than elsewhere in the Gulf ([Exhibit 7](#)).
 - Historical episodes in terms of the speed and extent of production recovery (partly reflecting the differences in causes of curtailments, and recognizing that today's shock is unprecedented) ([Exhibit 8](#)).
 - b. An average across external forecasts of Persian Gulf production (incl. from EIA, IEA) show recoveries as % of lost production of 70% after 3 months of reopening and of 88% after 6 months ([Exhibit 9](#)).
 - c. While not our baseline, the risk of large scarring to oil production capacity (we documented for several of the previous five largest oil supply shocks) may pick up if hostilities were to resume.

How Fast Can Gulf Production Recover After Reopening?

Exhibit 1: We Estimate 14.5mb/d of Persian Gulf Crude Production Curtailed in April, Primarily on Precautionary and Stock Management Measures

Estimated Gulf Crude Production in April 2026 (mb/d)				
	Estimated Crude Production Without War	Still Producing	Not Producing Because of Reported Physical Damage	Not Producing for Other Reasons
Total	25.4	11.0	0.3	14.2
Saudi Arabia	10.1	6.1	0.3	3.7
Iraq	4.4	1.4	0	3.0
Iran	3.5	1.0	0	2.5
UAE	3.5	0.8	0	2.7
Kuwait	2.6	1.4	0	1.2
Qatar	1.3	0.2	0	1.0

Under “Estimated Crude Production Without War” we report April 2026 production projections from our GS February 2026 balance. Argus reported on 12 April 2026 that Saudi Arabia’s Energy Ministry said repair work was ongoing at Khurais field, where production capacity had been reduced by about 0.3mb/d, following a strike on one of its East-West pipeline pumping stations. The Ministry said that full restoration would be announced once repairs were completed.

Source: IEA, Goldman Sachs Global Investment Research

Exhibit 2: Insufficient Reservoir Pressure/Flow Rates Following Forced Curtailments May Be a More Persistent Constraint on Production of Some Wells

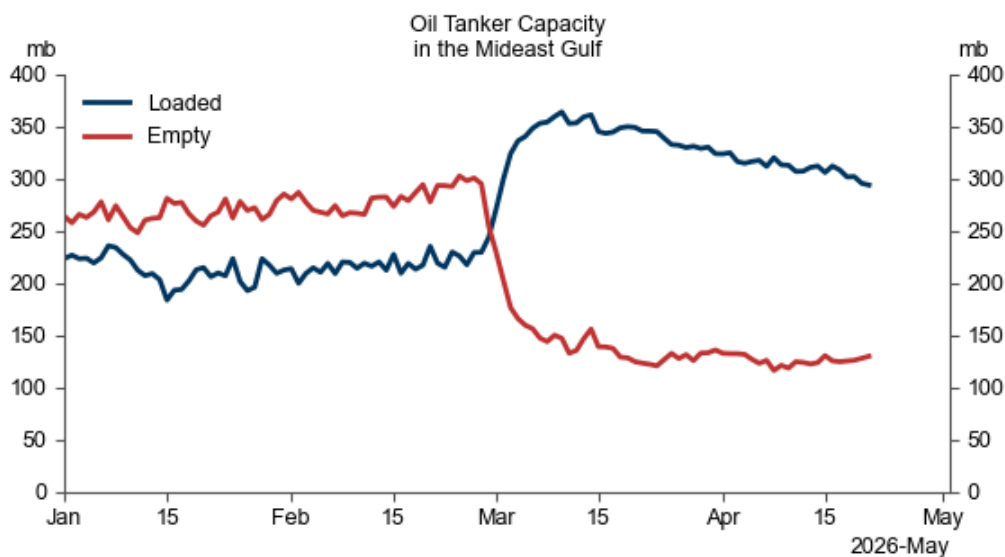
Prospects for Recovery in Upstream Oil Production			
Quote Number	Date	Speaker	Speed of Recovery
1	22-Apr	Girish K. Saligram (CEO of Weatherford International)	When you’ve also got production that shut in [...] it’s not a guarantee that you’re going to get back at the exact same flow rates .
2	21-Apr	Shannon Slocum (Executive Vice President and Chief Operating Officer of Halliburton)	I’d say the longer they [wells] get shut in, the more complex that gets .
3	21-Apr	Jeffrey A. Miller (Chairman of the Board, President and Chief Executive Officer of Halliburton)	The longer things are shut in, typically, the more complex they are to bring back on .
4	24-Mar	Sheikh Nawaf (CEO of Kuwait Petroleum Corporation)	Kuwait would take three to four months to return output to full production levels even if the war were to end today.
5	10-Mar	Amin Nasser (President and CEO of Saudi Aramco)	We can ramp up in days , and not weeks for sure and, to reach whatever, when we talk about, we have an ample spare capacity as well, and in certain areas, we use restriction on wells rather than shutdowns.

Source: Reuters, Saudi Aramco, Halliburton, Weatherford International, Goldman Sachs Global Investment Research

Production Process

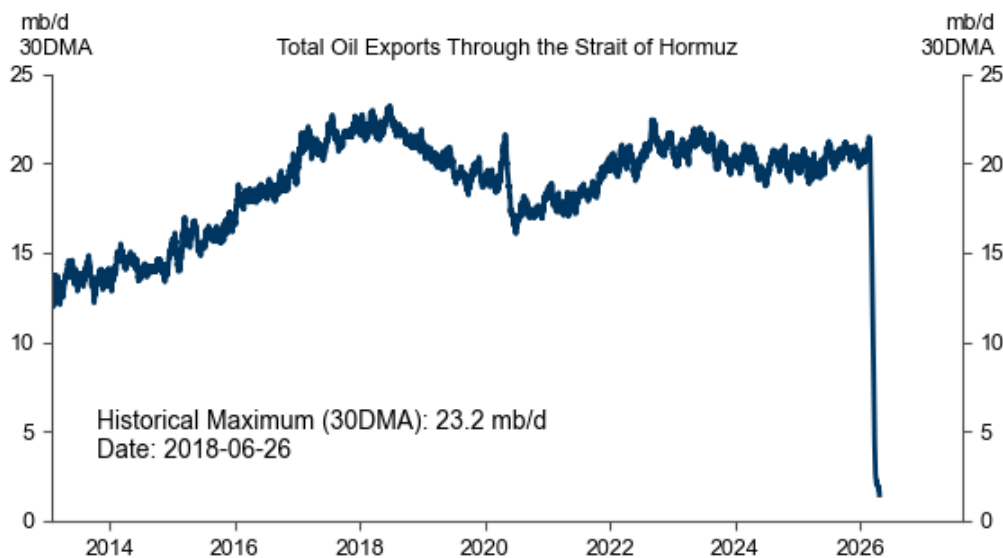
Transportation and Well Flow Rates Are Key Potential Constraints

Exhibit 3: We Estimate that Available Empty Tanker Capacity in the Persian Gulf Has Fallen by Around 130mb (Around 50%)



Source: Kpler, Goldman Sachs Global Investment Research

Exhibit 4: Oil Flows Through the Strait of Hormuz Exceeded Their Normal 20mb/d Pace by Over 3mb/d in mid-2018



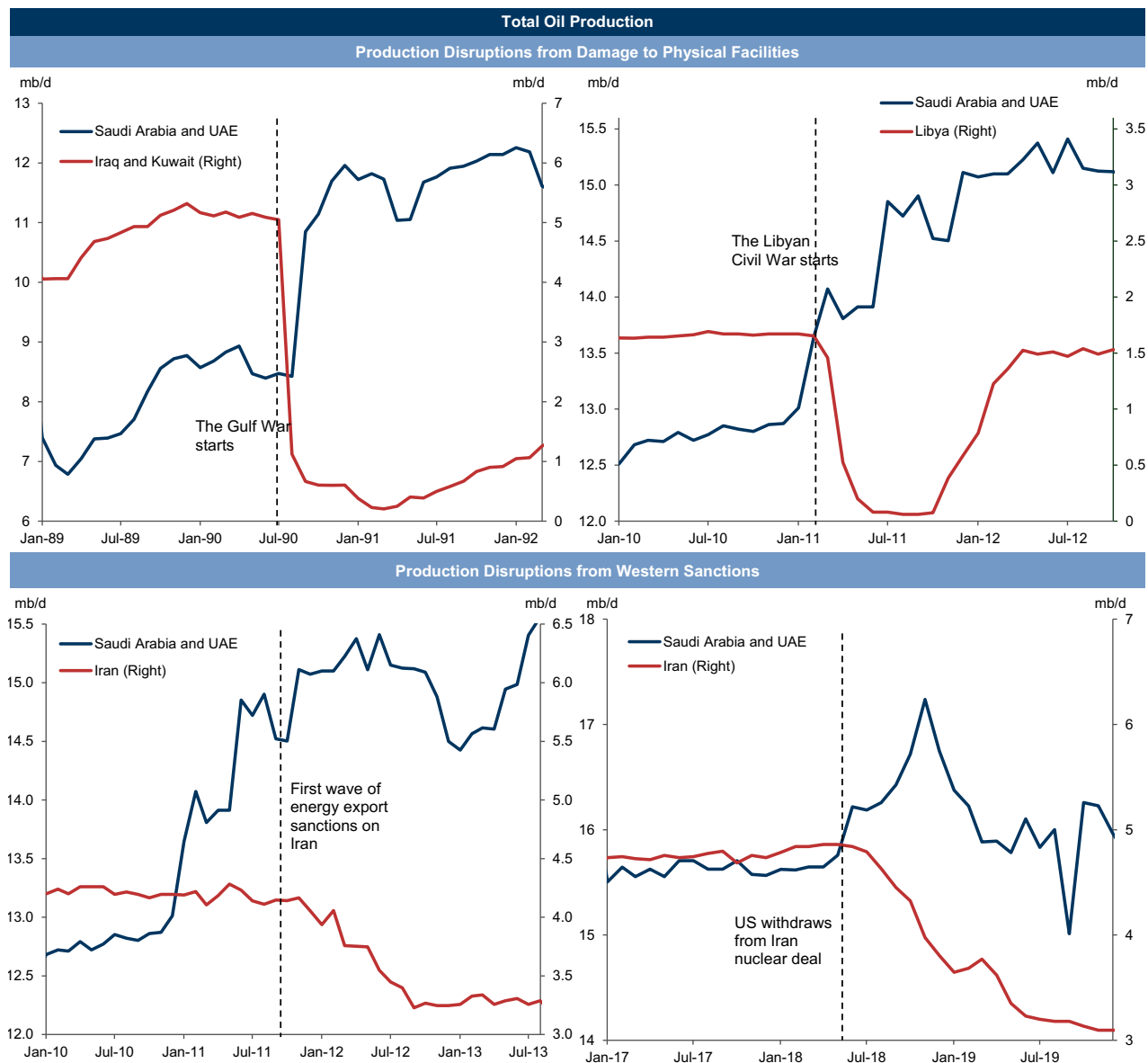
Oil exports includes crude, NGLs, and refined products.

Source: Kpler, Goldman Sachs Global Investment Research

The Case for a Solid Recovery

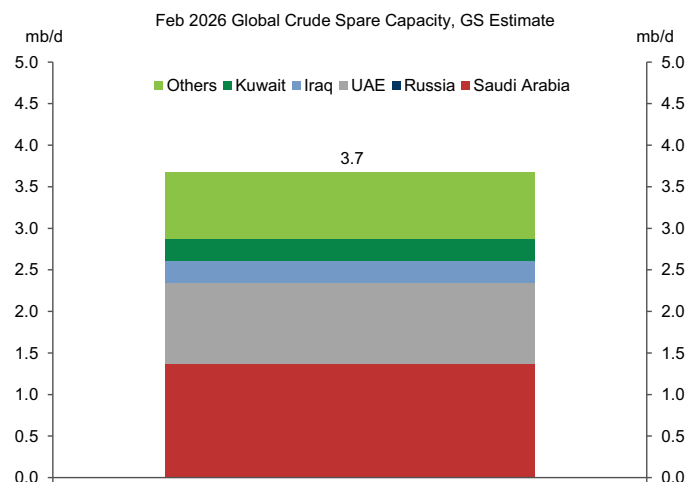
Core OPEC Market Stabilization

Exhibit 5: Saudi Arabia and UAE Production Historically Tends to (Eventually) Offset Much of Disrupted Supply



Source: IEA, Goldman Sachs Global Investment Research

Exhibit 6: We Estimate That Saudi Arabia and UAE Could Deploy Over 2mb/d of Spare Production vs. Pre-War Levels to Stabilize Markets After a Safe and Full Reopening

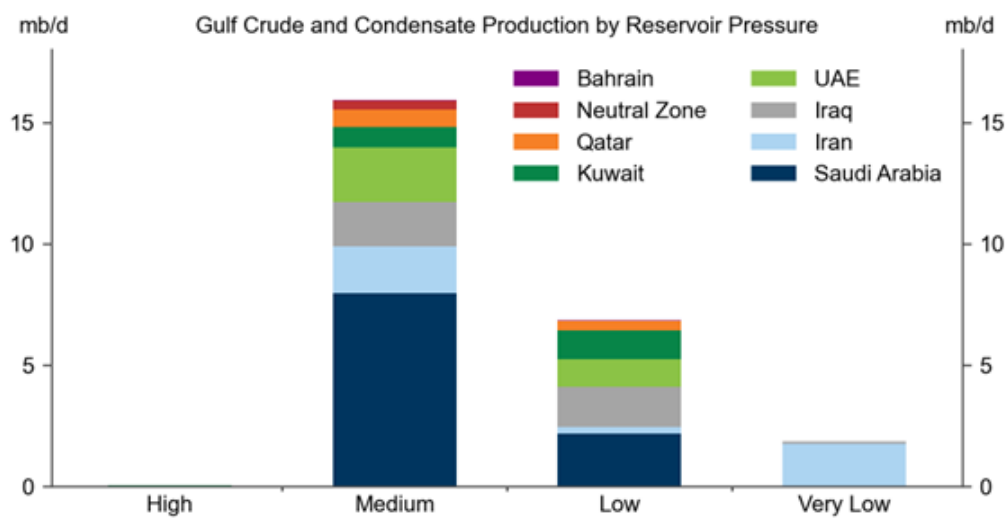


Source: IEA, EIA, Goldman Sachs Global Investment Research

Risks to Long-Run Production

Differences Across Fields and Countries

Exhibit 7: Most of Gulf Reservoirs Are Estimated to Have Medium Pressure, But Nearly Half of Iranian Reservoirs Are Estimated to Have Low Pressure



Source: Rystad Energy

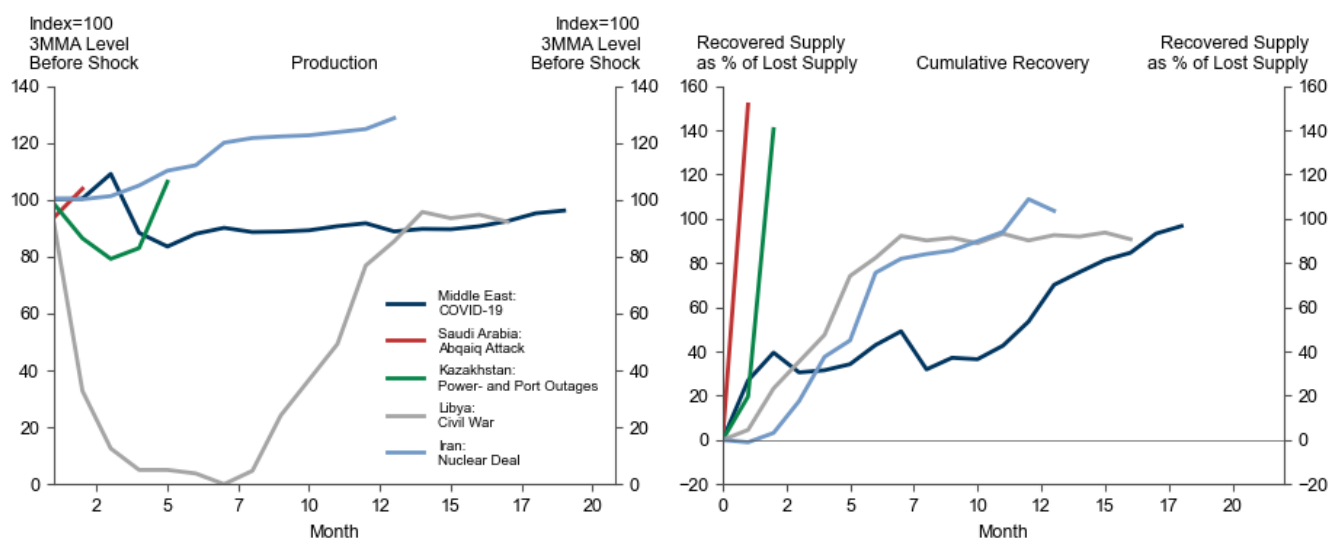
Production is a 2026 annual average projection. Pressure is not a sufficient statistic to predict workovers on wells. Other factors (e.g. sand granularity) play a role too.

Source: Rystad Energy, IEA, Goldman Sachs Global Investment Research

Differences Across Historical Production Recovery Episodes

Exhibit 8: While the Hormuz Shock Is Unprecedented, Our Analysis of Historical Production Recovery Episodes Shows Large Differences in Recovery Paces (Partly Reflecting Different Causes of Curtailments)

Differences Across Historical Crude Production Recovery Episodes



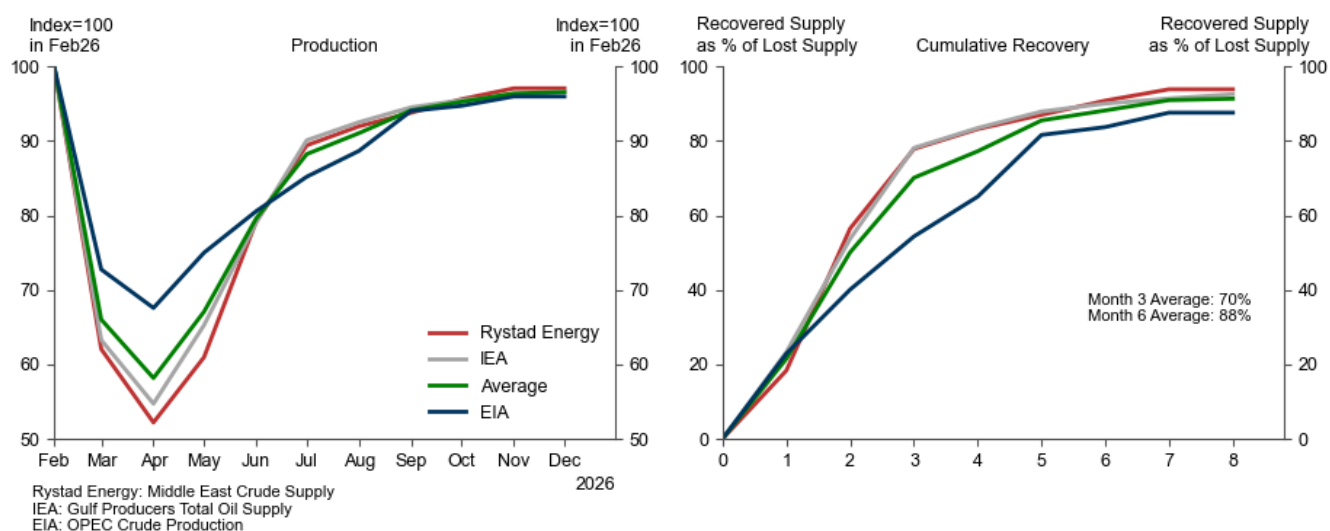
We consider crude production data.

Source: IEA, Goldman Sachs Global Investment Research

Forecasts From Energy Agencies

Exhibit 9: An average Across External Forecasts of Persian Gulf Production in 2026 Shows Recoveries as % of Lost Production of 70% After 3 Months of Reopening and of 88% After 6 Months

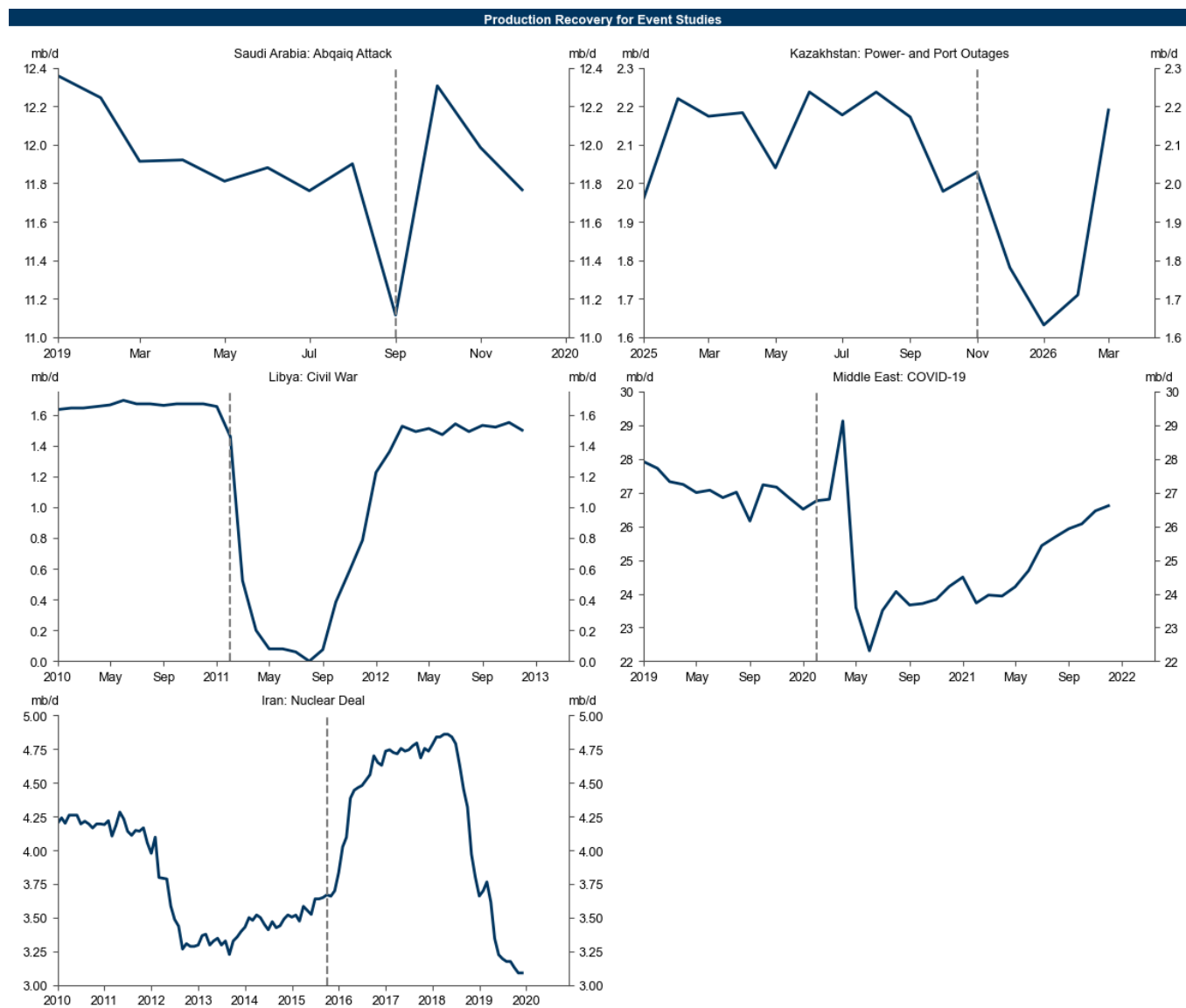
Forecasts Comparison Across Agencies



Source: Rystad Energy, IEA, EIA, Goldman Sachs Global Investment Research

Appendix

Exhibit 10: We Look at Five Historical Episodes of Production Disruptions and Recovery But Ramp-Up Patterns Differ Across the Shocks (Which All Differ in Cause and Nature)



Source: IEA, Goldman Sachs Global Investment Research

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